6



FOCUS

6 LEARNING PROTOCOLS

1. Shift into a State of Alertness and Focus
2. Use Mental Contrasting to Achieve a “State of Urgency” and Reach Your Goals
3. Embrace Errors to Improve Your Performance
4. Prioritize Sleep the Night After a Learning Bout
5. Incorporate Micro Rests into Your Learning Bouts
6. Increase Retention of New Skills and Information with Mental Rehearsal

Unofficial non-profit edition

Our ability to get and stay motivated, to focus, and to learn new things is at the heart of what it is to be successful in life. Not just in school and work, but in creative endeavors and athletics. Consider the effort it takes to study for a big test or remember the material needed for a big presentation, or even stick with a new approach to a painting or new musical instrument practice, or try to respond to a social situation in a new way.

The question then becomes, “How?” How can we get motivated when our energy or natural impulses pull us toward apathy, overthinking, or procrastination? How can we tap into heightened focus, on demand? How can we learn faster and more effectively?

It turns out these are states of mind and body that we can learn to deliberately access.

My second year of college is when I first started to discover this. I learned that motivation can happen automatically as the consequence of, say, being excited or intrigued by something but it can also be deliberately fostered.

As I mentioned a bit in the introduction, I went from being a good student in my early life to falling behind my peers in high school. I touched on why that happened, but I make no excuses. Those were ultimately my own choices and actions. And those challenges both in and out of school, and watching my peers face those challenges too—seeing some

soar and thrive while others struggled—would all become a blessing in many ways. It forced me to realize that no one was coming to save me and that I had to change things on my own. That realization scared me, but it's what motivated me to turn my life around. I never quite had the exact words for what went through my head at that moment until recently, when a very wise friend and mentor said something and I thought, "THAT! That's what I was hearing in my head! THAT!"

I can't say I came up with it, but it's something that I have heard in my head in fragments, over and over since I was nineteen. I still remind myself of it every day:

Losers take the negative things that happen to them and the feelings that come from things that suck… and they wallow and they use it for self- or outward destruction.

Winners however take whatever they feel—even if it sucks—and they transmute those feelings into actions that are good for themselves and for the world.

I chose to do something. And through that choice, I learned that while motivation can result in action, it's possible for action to result in motivation too.

As a sophomore in college, I decided to study as if my life depended on it. I didn't care if I liked the subject. I decided it made no difference, that eventually I would find the right

path if I kept moving forward but I was going to move forward no matter what. Sure enough, my motivation to learn grew more and more, and morphed into a desire to get a PhD in a topic I fell in love with, to run a lab, to teach, and to build a career in science. Somewhere early along the way that fear I described dissipated. My love of certain topics and lab skills grew, but it often came after I had forced myself to learn them. Sure, sometimes it was there prior—I've always had a love for biology for example—but often it was only after I immersed myself in a subject that I truly fell in love with it. The hardest courses I took in college turned out to be the ones I loved the most, and years later those subjects would be among the major focuses of my lab. The most useful thing I learned back then was that I realized that I could access the state of urgency that we call motivation whether I loved something or not. Reason and persistence, not excitement, anchored my effort.

And the act of being able to access a state of motivation gave me confidence to tackle what I wanted to go after or what was placed in front of me. I wasn't a grade-chasing automaton; I had a sense of agency over my life and that kept me going.

Just as my motivation grew, so too did my ability to focus and learn. At first, I had to take drastic measures to force myself to do both. While studying biology in university, I

would set a timer for several hours and force myself to study without breaks until the timer went off. Having been so checked out in high school, I had a lot of catching up to do. This was extremely difficult at first. Long bouts of studying required a level of focus I did not have back then. But over time, my ability to focus got better—and kept getting better.

I later came to discover that the early stages of focus and learning are supposed to include agitation. That feeling comes from the shift in your brain and body's neurochemical milieu that sets the stage for learning to take place. You are asking your brain to change, and your brain doesn't want to change. Changing is very energetically demanding for brains. This uncomfortable feeling is something I often refer to as "limbic friction." It's not an official scientific term, but it refers to a state of elevated autonomic arousal where you're funneling into focus, and it takes constant or near-constant effort to get there and stay there. It's a mental edge of sorts. Once I understood that this limbic friction was not only normal, but essential, focus and learning became easier. I started seeking it out. And it felt good. I share this backstory simply to help illustrate that motivation, focus, and learning are states of mind and body. They have neural and hormonal underpinnings, much in the same way that REM sleep has its neural and hormonal underpinnings, as does stress, and so on. And that's good news, because it

makes what might otherwise be nebulous, clear. We know what to aim for.

This is a physiology-first approach to focus, motivation, and learning. It's rooted in how we work, which is why the approach works.

In this chapter you'll learn how to enhance your ability to focus by leveraging that. You'll also learn a powerful tool for "getting motivated" that is unlike all others you've likely seen before. And you'll finally get the truth about neuroplasticity—the rewiring of neural circuits that underlies all forms of learning (and forgetting), and that is often referred to simply as plasticity. Most everyone has heard of plasticity but honestly, most of the ways I've seen it discussed publicly are flat-out wrong.

Perhaps you've heard that "Every experience you have changes your brain." That is not true. Or "It takes twenty-one days to form a new habit." Wrong again. It can take a few days or a year, based on your level of motivation. Perhaps you've heard that "Your brain makes new neurons for your entire life." (The official term for that is neurogenesis.) Alas, barely any new neurons are added to the brain after puberty except for in your olfactory system, which allows you to smell, and those are mostly to replace neurons you've lost. Starting to get depressed? Well don't, because despite these

myths, we all possess a tremendous ability to rewire our brains and to learn at any age.

The key is to understand the type of plasticity you're after. What we're talking about here—the rewiring of our brains that occurs as part of learning at any age—is a process that involves a specific set of steps that must be completed in a specific sequence for learning to occur.

I call these steps self-directed adaptive plasticity (SDAP). Mastering these steps opens the doors to learning pretty much anything you set your mind to. And that's what this chapter is about.

The Steps and Sequence of SDAP

Let's start with an overview of how SDAP occurs:

Get alert.

SDAP requires you to focus on the material at hand. Alertness sets the stage for focus; it's a precursor. I'll explain all this in the pages ahead.

So get alert and…

Get and stay focused.

Focused on what? On implementing a specific strategy during your learning bout.

As you go…

Maintain a state of urgency.

Your internal state really matters as you learn. You can't be too relaxed, but you can't be stressed either. I define a state of urgency as simply a state of persistent motivation. It's the state to aim for because it has you stay alert but not too alert and…

Embrace errors.

Errors that arise from genuine attempts to do something are key signals for triggering plasticity. Your nervous system is stimulated to rewire largely based on what you fail to do, not what you do correctly.

Rest.

Sleep is when neural rewiring actually occurs, especially during REM sleep the first night after a learning bout. There are other ways you can incorporate rest—like non-sleep deep rest (NSDR) and short breaks—to improve performance.

Repeat.

Consolidate newly learned material or skills through repeat performance, either through physical repetition or taking yourself through the material or skill using your imagination (seriously). Mental self-testing is especially effective for reinforcing newly formed brain connections.

Each of these six steps is valuable in its own right, and can be applied to various areas of life. Thus, although this chapter is framed in the context of plasticity, each of the protocols has other important applications too. I'll explain more as we go.

Types of Neuroplasticity

Before we explore each step of SDAP, let's get clear on what we're talking about when we talk about plasticity.

All sustained behavioral and cognitive change is the result of plasticity. These changes come in two broad types: adaptive and maladaptive. Adaptive changes are desirable—this is the plasticity that accompanies learning a new language or a new artistic or athletic skill. Maladaptive changes are the ones that make life harder. Consider when a brain injury renders someone unable to produce coherent speech, or a traumatic event leaves someone stricken with PTSD. That's still plasticity but obviously not the sort you want.

These two types of plasticity happen in different ways. Maladaptive plasticity, as in the brain injury I described above, tends to occur fast and is hard to undo. Adaptive plasticity, like learning that new language, takes more effort, and reverts somewhat easily, at least at first. I realize that's not convenient but the sooner we accept that reality, the further along we can all be in our efforts to learn new things.

Understanding why and how maladaptive brain changes occur will help illustrate how to get the adaptive brain changes that you want, and make them happen more quickly. Psychologically traumatic events trigger astonishingly fast and stable changes to nervous system wiring.[1] This rewiring happens not just in the brain but in the patterns of brain-to-body and body-to-brain connections, and hormone-control pathways.[2] The duration of a traumatic event can be very brief—no more than a few seconds—yet bring about persistent or recurring anxiety, sleep disturbances, and physical challenges, like chronic pain,[3] rashes,[4] gut issues,[5] and more, that last for years.

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[1] Geraldine Gvozdanovic et al., "Experimental Trauma Rapidly Modifies Functional Connectivity," Brain Imaging and Behavior 15 (2021): 2017–30, https://doi.org/10.1007/s11682-020-00396-2; Xin Wang et al., "Preliminary Study of Acute Changes in Emotion Processing in Trauma Survivors with PTSD Symptoms," PLOS One 11, no. 7 (2016): e0159065, https://doi.org/10.1371/journal.pone.0159065.

[2] Felicia A. Hardi et al., "Trauma-Predictive Brain Network Connectivity Adaptively Responds to Mild Acute Stress," Proceedings of the National Academy of Sciences 122, no. 31 (2025): e2505965122, https://doi.org/10.1073/pnas.2505965122; Dongju Seo et al., "Limbic Response to Stress Linking Life Trauma and Hypothalamus-Pituitary-Adrenal Axis Function," Psychoneuroendocrinology 99 (January 2019): 38–46, https://doi.org/10.1016/j.psyneuen.2018.08.023.

[3] Mark A. Lumley et al., "Trauma Matters: Psychological Interventions for Comorbid Psychosocial Trauma and Chronic Pain," Pain 163, no. 4 (2022): 599–603, https://doi.org/10.1097/j.pain.0000000000002425.

[4] Neslihan Cansel et al., "The Role of Childhood Trauma in Patients with Chronic Urticaria," Northern Clinics of Istanbul 10, no. 1 (2023): 79–86, https://doi.org/10.14744/nci.2021.10170.

[5] Shanlin Ke et al., "The Emerging Role of the Gut Microbiome in Posttraumatic Stress Disorder," Brain, Behavior, and Immunity 114 (November 2023): 360–70, https://doi.org/10.1016/j.bbi.2023.09.005.

This is essentially one trial learning. Something happens and we are (seemingly) forever changed. The evolutionary logic is straightforward: One trial learning from traumatic events helps ensure we avoid similar circumstances in the future. It keeps us safe.

How one trial learning happens brings us back to an analogy discussed in chapter 3. Remember the autonomic seesaw? A stress-inducing event or thought causes your autonomic nervous system to increase your sympathetic tone, which makes you alert. That sharpens your focus. If sympathetic tone is too high, however, it feels like anxiety or panic.

When we experience a traumatic event, our brains quickly trigger the release of a lot of adrenaline into our bodies and a lot of norepinephrine in our brains.[6] Together, these two neuromodulators create all the necessary conditions for plasticity to occur. We're not talking about a bit of normal life stress. We're talking about something you perceive as really scary that causes a large and rapid shift in the neurochemical milieu of the body and brain. This shift during a traumatic event may sound like a good thing, because it helps you move away from the person or thing stressing you out, but in the long run many times it's not. If the stress response is too big, it triggers the rapid modification of multiple brain circuitries all at once and can

[6] Brianna Chu et al., "Physiology, Stress Reaction," StatPearls, last updated May 7, 2024, https://www.ncbi.nlm.nih.gov/books/NBK541120/.

result in an unhealthily low stress threshold going forward.[7] In some cases, one's stress threshold becomes set so low that anxiety and panic arise from innocuous events of everyday life. Undoing the kind of rewiring that results from traumatic events takes dedicated effort. Fortunately, it is possible. I discuss one way to do that in the next chapter.

The conditions experienced during this kind of one-trial learning are also at play in SDAP. "Uh-oh," you may be thinking, "is he going to suggest that I get stressed out to learn?" No. In fact, if you're too stressed, and your sympathetic tone is too elevated, you won't be able to focus, and you absolutely need to focus in order to learn. But you do need to be alert. My point here is that a shift in neurochemical milieu is crucial to achieving that state—alert but calm. Not stressed.

Conditions for Neuroplasticity

SDAP is all about desired forms of learning: a new musical instrument, a new language, a new physical skill, a new healthy habit, et cetera. For SDAP to occur, you must focus intensely on what you are trying to learn. To focus intensely on what you are trying to learn, you have to be alert. And to

[7] Amy F. T. Arnsten, "Stress Signalling Pathways That Impair Prefrontal Cortex Structure and Function," Nature Reviews Neuroscience 10, no. 6 (2009): 410–22, https://doi.org/10.1038/nrn2648.

be alert, your autonomic nervous system has to be in a somewhat sympathetic-dominant state.

That seesaw shift in the nervous system signals to the relevant circuits in your brain and body that what is happening is unique and important, and that they need to change to become better equipped for similar circumstances in the future. Another useful way to think about it is that your brain must get the message that there is a gap between its current configuration and the demands you're facing—a deficit of sorts that it needs to overcome. The deficit is resolved by making new brain connections.

Now of course your neurons don't talk to each other in words. They don't say, "This is important, and we have a deficit between where we're at and where we need to be, so we need to change." Neurons communicate in the language of chemicals and electricity. So this message is communicated electrochemically, as a shift in your brain and body's neurochemical milieu that in turn alters the amount of firing in the brain circuits that need to change for the desired learning to occur. And when it comes to shifting neurochemical milieus—that is largely the job of neuromodulators.

The neuromodulators dopamine, serotonin, adrenaline, norepinephrine, and acetylcholine each serve different purposes, depending on where they are released in your

brain or body.[8] It's incorrect to say that dopamine alone drives motivation, for instance, or that serotonin controls mood. It's far more varied and nuanced than that. In terms of plasticity, all these neuromodulators can alter the level of electrochemical activity in the circuits I just described. In other words, these neuromodulators can change the amount and the volume that neurons "talk" to each other.

In fact, in the context of plasticity, you can think of a neuromodulator as a microphone lowered over a specific conversation in a crowded stadium. Like a stadium on game day, brains are noisy. They are busy carrying out all kinds of everyday electrochemical duties. The "microphone" makes a conversation louder against the background noise. This is one common way plasticity occurs. The faster and greater the shift in the activity of a given circuit relative to its previous levels, and relative to the background (more electrochemically active), the more likely that circuit will alter its structure and connections—in other words, the more likely plasticity of the circuit will occur. It's also true that if neural circuits are quieter (less electrochemically active) against the chatter of usual background brain activity, they can weaken.

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[8] Michael C. Avery and Jeffrey L. Krichmar, "Neuromodulatory Systems and Their Interactions: A Review of Models, Theories, and Experiments," Frontiers in Neural Circuits 11 (December 21, 2017): 108, https://doi.org/10.3389/fncir.2017.00108.

Different contexts spike the release of different neuromodulators alone or in combination, which amplifies the conversation of the neural circuits active in that context. For example, dopamine is a major neuromodulator released in response to novelty and surprise, which facilitates plasticity for learning and memory.[9] And acetylcholine is released at high levels under conditions of heightened focus, but not all over the brain; rather, it's released specifically at the sites in neural circuits that are active.[10]

Regardless of which neuromodulator or combination of neuromodulators is released, what happens next is largely the same. Since the brain is homeostatic—that is, it seeks balance—rather than maintain high levels of activity, the neurons that are part of a given conversation modify themselves so they become more easily active should that context occur again. They may add more receptors to render themselves capable of capturing more neurotransmitters or they may sprout a tiny neurite, an axonal (output) or dendritic (input) branch that becomes part of a new set of synaptic connections with other nearby neurons.[11] All this

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[9] Adrian J. Duszkiewicz et al., "Novelty and Dopaminergic Modulation of Memory Persistence: A Tale of Two Systems," Trends in Neurosciences 42, no. 2 (2019): 102–14, https://doi.org/10.1016/j.tins.2018.10.002.

[10] Vinay Parikh and Debra A. Bangasser, "Cholinergic Signaling Dynamics and Cognitive Control of Attention," in Behavioral Pharmacology of the Cholinergic System, ed. M. Shoaib and T. Wallace (Springer Nature Switzerland, 2020), 71–87, https://doi.org/10.1007/7854_2020_133.

[11] Liqun Luo, Principles of Neurobiology (CRC Press, 2020).

changes the threshold for activation of that circuit going forward, and on and on.[12] Voila! That is learning-driven plasticity. Other forms of plasticity involve similar cellular and molecular changes but the conditions that create them are different.

SDAP is about creating this situation of specific neural circuits rewiring in specific ways on purpose. The steps involved in this process are designed to help you do just that.

Where to Begin

Above, I noted that the first step of SDAP is getting alert and the second is getting focused. These two work together in a positive feedback loop.

The cool thing is even the mere act of trying to focus shifts us into a slightly more elevated sympathetic tone, which causes our levels of alertness to increase, which then causes activity in the circuits for focusing to increase, which further causes levels of alertness to, you guessed it, increase. For those who struggle with focus, take solace in the fact that the circuits involved with focus are amenable to plasticity too. As you move through the SDAP process described in this chapter, and practice the various protocols here, you'll learn new things, but you'll also get better at focusing, which will help you to learn more new things, and so on.

[12] Matt Puderbaugh and Prabhu D. Emmady, "Neuroplasticity," StatPearls, last updated May 1, 2023, https://www.ncbi.nlm.nih.gov/books/NBK557811/.

There are multiple ways to create the increase in alertness and focus needed for SDAP. I'll cover all your options—behavioral, supplement based, pharmacologic—below.

But remember this: Alertness and focus are necessary but not sufficient for plasticity and learning. To trigger learning, you're also going to have to make errors from genuine attempts, what I refer to as repetitions, or reps. Not deliberate errors, and not half-baked efforts, but rather, errors that emerge from truly exerting yourself to learn new information, new movements, even new modes of thinking. Consider this: If your nervous system can already generate the activity patterns that enable you to do a certain task—perform a dance step, for example, or order coffee in a foreign language—why would it change? It doesn't need to! You're already equipped with the neural circuits to accomplish that task. You need to try, and fail, and try again. This is tough to accept, especially for perfectionists, but all who embrace it thrive. Especially those who come to recognize limbic friction, the slight agitation that accompanies effective learning, as a state to seek, not shy away from.

The importance of making mistakes and trying again doesn't mean you have to toil endlessly. SDAP can be enjoyable. You'll likely experience some limbic friction but guess

what? Once you learn to expect it, and you realize that it helps not hinders your ability to learn, you too will be seeking it out.

You may be wondering “Why all these conditions for plasticity? Isn’t there a plasticity pill?” No. As I mentioned earlier in this introduction, plasticity is energetically expensive. Strengthening and reshaping neural circuits requires a lot of ATP, the energy our cells need to function.

And this process applies to everyone. It’s true that when it comes to neuroplasticity, children have it easier than adults. Their brains are more malleable because almost every experience is novel to them. In fact, many aspects of developmental plasticity emerge with no effort or need to deliberately focus at all. Good examples include the way that a baby’s visual and auditory systems are shaped by their early visual and sound experiences. Those things just happen. The automaticity of plasticity in our youth isn’t always a blessing however. If a child has a temporary eye injury or if a child’s eyes are misaligned due to an imbalance in the muscles that help move the eyes, it can cause longstanding maladaptive changes in brain wiring.

Yet even at an early age, many forms of plasticity still require focus. Cognitive skills like language and motor skills like walking or eating with utensils all take deliberate focus to learn. In that sense, the process for motor and cognitive

skill learning is inherently the same for all ages. Your states of mind and body, alongside the agitation you experience from errors while making genuine attempts to learn, are what ensure that your brain gets the message: This thing happening right now is unique and important; we're going to have to rewire.

As You Proceed

An important point for anyone wishing to establish new brain patterns is that following each step of the SDAP process in its proper sequence is more important than any one tool listed here. This will be an important reframing for a lot of people.

I mentioned that there wasn't a pill for plasticity, but people love to think about what they can take to increase neuromodulator levels to heighten focus. Adderall, energy drinks, nicotine, modafinil, microdoses of psilocybin, et cetera, are used by many people working in business, tech, sports performance, finance circles, and beyond to try to focus and learn better. The use of such stimulants is widespread on college campuses and among high school students in the US,[13] and millions of adults use them daily too.

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[13] Joshua P. Rising et al., "United States Trends in Non-Prescribed Use of Adderall and Ritalin: Population Assessment of Tobacco and Health (PATH) Study Estimates from 2013 to 2022," Addiction 120, no. 11 (2025): 2348–54, https://doi.org/10.1111/add.70123; Justine W. Welsh et al., "Substance Use

Their widespread use merely speaks to how hard most people find it to focus intensely enough and push through limbic friction. I'm not judging anyone. For example, I, like most adults in the US,[14] use caffeine daily to increase focus. I occasionally rely on other pharmacologic tools for focus too. In this chapter, I'll cover the popular as well as the rarely discussed ones.

That all being said, the main focus here is behavioral tools. They may not seem as sexy, but many rival the pharmacologic tools.

It truly doesn't matter what you use to get alert and focused, as long as it's safe. If you opt to use nicotine or modafinil or meditation or good ol' sleep and hydration, those obviously have different safety profiles. You can use white noise or various sound tones.[15] These all have data to support them in the context of focus and learning. You can cycle different tools in and out, by the day or by the month. The point is:

Among College Students," Focus 17, no. 2 (2019): 117–27, https://doi.org/10.1176/appi.focus.20180037; Christian J. Teter et al., "Nonmedical Use of Prescription Stimulants Among US High School Students to Help Study: Results from a National Survey," Journal of Pharmacy Practice 33, no. 1 (2020): 38–47, https://doi.org/10.1177/0897190018783887.

[14] Diane C. Mitchell et al., "An Update on Beverage Consumption Patterns and Caffeine Intakes in a Representative Sample of the US Population," Food and Chemical Toxicology 196 (February 2025): 115237, https://doi.org/10.1016/j.fct.2025.115237.

[15] Mohamad Awada et al., "Cognitive Performance, Creativity and Stress Levels of Neurotypical Young Adults Under Different White Noise Levels," Scientific Reports 12, article no. 14566 (2022), https://doi.org/10.1038/s41598-022-18862-w; Anthony J. Angwin et al., "White

You have choices. Think of them as tools in your SDAP arsenal.

Stick with It

We know a lot more about plasticity now than we did twenty-five or even just five years ago. Prior to the 2000s the widely held belief—even among neuroscience researchers—was that the opportunity for adaptive plasticity mostly closes after age twenty-five.[16]

That idea is now dead. We know people can rewire their brains and learn throughout their lifespans.[17] Also mostly dead is the idea that learning-associated plasticity involves the addition of new neurons. Plasticity in humans is almost

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Noise Enhances New-Word Learning in Healthy Adults," Scientific Reports 7, article no. 13045 (2017), https://doi.org/10.1038/s41598-017-13383-3; Elza Othman et al., "Low Intensity White Noise Improves Performance in Auditory Working Memory Task: An fMRI Study," Heliyon 5, no. 9 (2019): e02444, https://doi.org/10.1016/j.heliyon.2019.e02444.

[16] Liqun Luo, Principles of Neurobiology (CRC Press, 2020); Patrícia Marzola et al., "Exploring the Role of Neuroplasticity in Development, Aging, and Neurodegeneration," Brain Sciences 13, no. 12 (2023), https://doi.org/10.3390/brainsci13121610; Alvaro Pascual-Leone et al., "The Plastic Human Brain Cortex," Annual Review of Neuroscience 28 (July 2005): 377–401, https://doi.org/10.1146/annurev.neuro.27.070203.144216; Pasko Rakic, "Limits of Neurogenesis in Primates," Science 227, no. 4690 (1985): 1054–56, https://doi.org/10.1126/science.3975601; Beatrix P. Rubin, "Changing Brains: The Emergence of the Field of Adult Neurogenesis," BioSocieties 4 (December 2009): 407–24, https://doi.org/10.1017/S1745855209990330; Michael M. Merzenich et al., "Brain Plasticity-Based Therapeutics," Frontiers in Human Neuroscience 8 (2014): 385, https://doi.org/10.3389/fnhum.2014.00385.

[17] Marzola et al., "Exploring the Role of Neuroplasticity"; Merzenich et al., "Brain Plasticity-Based Therapeutics."

exclusively the consequence of changes to existing neural circuits, not adding new neurons to those circuits.[18] And yet, the results you can achieve when you incorporate the best of what we know about SDAP can be extraordinary.

Here is an overview of SDAP, once more:

- Get alert.
- Get and stay focused.
- Maintain a state of urgency.
- Embrace errors that arise from genuine attempts.
- Rest.
- Repeat your performance.

I've listed these again because the steps and the sequence are non-negotiable. Not because I say so, but because that's how our neuroplasticity works. There are things you can do to accelerate the process: all the tools I mentioned above (and more), along with micro-pauses, post-learning-bout NSDR, adjusted sleep protocols, and hypnosis. I cover all these in the pages that follow.

I began this chapter with this basic process because it can be applied to learning anything.

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[18] Chiara La Rosa et al., "Brain Structural Plasticity: From Adult Neurogenesis to Immature Neurons," Frontiers in Neuroscience 14 (2020): 75, https://doi.org/10.3389/fnins.2020.00075.

Similar to exercise, SDAP is not easy. But with time, each step becomes familiar. If you think of SDAP this way—as a process that consistently leads to a desired end—it can become a pleasure.



LEARNING PROTOCOL

Shift into a State of Alertness and Focus

WHAT TO DO

- Getting alert and focused is the first step in SDAP. You have to be in a state of elevated autonomic (sympathetic) arousal, but not stressed.
- Use tools to help you achieve this state. Such tools are many and varied, and fall into three general categories: Foundational
- Behavioral
- Pharmacological
- Put your phone away. You may think you have an issue with getting focused, but it's likely that you are distracted by something in your surroundings—the smartphone being the major culprit. Studies show that even just having your phone in the same room in a bag behind or

under your chair reduces your ability to focus on cognitive tasks and/or increases the energy that it takes.[19]

HOW IT WORKS

You might be thinking "Why would anyone need a protocol for focus?"

Fair question. Most people can easily get (and stay) focused on texting, scrolling social media, playing a video game, or watching their favorite sports team or TV show for an hour or more, with no protocol required.

But when it comes to SDAP—pushing your brain to understand or do that which it currently cannot understand or do—focusing is hard for most people, young or old.[20] One recent, albeit small, study of fifteen people engaged in a work-related task found that their focus wavered every 3.5 minutes, on average.[21] The latest data say that most people

[19] Jeanette Skowronek et al., "The Mere Presence of a Smartphone Reduces Basal Attentional Performance," Scientific Reports 13, article no. 9363 (2023), https://doi.org/10.1038/s41598-023-36256-4.

[20] Adrian M. Haith and John W. Krakauer, "The Multiple Effects of Practice: Skill, Habit and Reduced Cognitive Load," Current Opinion in Behavioral Sciences 20 (April 2018): 196–201, https://doi.org/10.1016/j.cobeha.2018.01.015.

[21] Dinara Talypova et al., "Are Six Minutes of Focus Enough? An Exploratory Study of Multitasking Patterns in Workplace Environments," Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work, CHIWORK 2025, Amsterdam, Netherlands, June 23–25, 2025.

switch tasks every couple of minutes, or even less.[22] And that number drops further when people are trying to focus on challenging material. That is why I refer to a dedicated session of trying to learn something as a learning bout. It is a fight—with the material you're trying to learn, and with yourself as you try to stave off frustration and resist the temptation to quit.

Yet it's in that state of mild agitation that plasticity is sparked. But only if we focus intensely on what it is we are trying to learn.

We can all take solace in the fact that getting and staying focused has always been hard. History shows that we humans have long sought tools to make it easier. In the early 1900s an inventor by the name of Hugo Gernsback designed the Isolator—a metal helmet that covered the entire head and face, with eyeholes and a breathing tube cut into the front.[23] It resembled the helmet of an old-school scuba diving suit. The rationale was simple: Block peripheral vision and outside noise to help people focus. The Isolator did not become popular, but 117 years later it inspired the Helmfon, an oversized motorcycle helmet designed to be worn while seated at a desk, created by industrial design company

[22] David Adam, "Are Attention Spans Really Shrinking? What the Science Says," Nature 653 (2026): 20–22, https://doi.org/10.1038/d41586-026-01407-w.

[23] Hugo Gernsback, "The Isolator," Science and Invention 13, no. 3 (July 1925).

Hochu rayu.[24] The rationale was the same, all that time later: Block out peripheral vision and outside noise to help people focus. The Helmfon, as you might imagine, also did not gain traction. Even if it had, the advent of social media and other online activities essentially moved the outside noise of the entire world directly into one's central field of view. So much for helmets.

Other long-used tools include non-amphetamine stimulants such as caffeine, which blocks adenosine receptors in the brain that would otherwise trigger sleepiness, and nicotine, which triggers the release of several neuromodulators discussed in the introduction to this chapter. The use of caffeine in the form of tea or coffee dates back more than two thousand years. But coffee and tea didn't become popular in Europe until the seventeenth century, the result of invasions and trade.[25] Today, approximately 90 percent of the world's adults—that's estimated as more than five billion people (!)—drink caffeine every single day. Later I'll explain how to use caffeine specifically to enhance focus, but meanwhile those stats reveal that caffeine is the most popular drug on the planet by a huge margin. (If you're

[24] Helmfon, Hochu-rayu, n.d., https://hochurayu.com/product_design/helmfon/.

[25] Zofia Kobylińska et al., "Coffee as a Source of Antioxidants and an Elixir of Youth," Antioxidants 14, no. 3 (2025): 285, https://doi.org/10.3390/antiox14030285; "China-England Tea Trade," EBSCO, 2022, https://www.ebsco.com/research-starters/history/china-england-tea-trade.

wondering, alcohol and nicotine are in a distant second and third place, respectively.)

Nicotine, mostly in oral forms like pouches, gum, and mints, has made a big comeback as a focus aid in recent years. I recently gave a talk to several thousand undergraduate men and women at an East Coast university and asked the auditorium for a show of hands of "who in here takes nicotine daily?" About 30 percent of them raised their hands. (In a more private setting—a smaller group of about fifty male students—about two-thirds said they use nicotine two or more times per day every day.) This is obviously not a controlled study, nor did I count each hand, but even if these are artificially high estimates, it's still pretty shocking. Or is it? Nicotine is an unusual stimulant because it makes people alert but calm—a highly desirable state for learning. Unless smoked, vaped, dipped, or snuffed, nicotine does not cause cancer, but it is highly addictive and it raises blood pressure.[26] Later, I'll explain how, assuming you're okay with the risks, nicotine can be used to enhance focus. There are a few safer alternatives, some of which work through the same mechanisms.

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[26] Gabriela F. Gil et al., "Health Effects Associated with Chewing Tobacco: A Burden of Proof Study," Nature Communications 15, no. 1 (February 2024): 1082, https://doi.org/10.1038/s41467-024-45074-9; Alpana K. Gupta et al., "Oral Cancer Risk Assessment for Different Types of Smokeless Tobacco Products Sold Worldwide: A Review of Reviews and Meta-Analyses," Cancer Prevention Research 15, no. 11 (September 2022): 733–46, https://doi.org/10.1158/1940-6207.capr-21-0567.

And there are the amphetamines. First synthesized in Romania in the late 1880s, amphetamines are compounds that increase levels of some of the key neuromodulators discussed in the introduction to this chapter, like norepinephrine and dopamine.[27] These stimulants were initially marketed in the 1930s as a treatment for narcolepsy, but soon hit the college scene when research showed a link with increased scores on intelligence tests.[28] They were also investigated for the treatment of attention deficit hyperactivity disorder (ADHD) and as appetite suppressants starting even back then.[29] A decade later, both German and Allied forces used amphetamines during World War II to stay awake and alert.[30] Recreational use of amphetamines exploded in general popularity starting in the 1950s, partly

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[27] S. M. Berman et al., "Potential Adverse Effects of Amphetamine Treatment on Brain and Behavior: A Review," Molecular Psychiatry 14, no. 2 (2009): 123–42, https://doi.org/10.1038/mp.2008.90; Erratum, Berman et al., "Potential Adverse Effects of Amphetamine Treatment," Molecular Psychiatry 15, no. 11 (2010): 1121, https://doi.org/10.1038/mp.2010.39.

[28] David J. Heal et al., "Amphetamine, Past and Present—a Pharmacological and Clinical Perspective," Journal of Psychopharmacology 27, no. 6 (2013): 479–96, https://doi.org/10.1177/0269881113482532.

[29] Madeleine P. Strohl, "Bradley's Benzedrine Studies on Children with Behavioral Disorders," Yale Journal of Biology and Medicine 84, no. 1 (2011): 27–33, https://pubmed.ncbi.nlm.nih.gov/21451781/; Stanley C. Harris et al., "The Mechanism of Amphetamine-Induced Loss of Weight: A Consideration of the Theory of Hunger and Appetite," JAMA 134, no. 17 (1947): 1468–75, https://doi.org/10.1001/jama.1947.02880340022005.

[30] Nicolas Rasmussen, "America's First Amphetamine Epidemic 1929–1971: A Quantitative and Qualitative Retrospective with Implications for the Present," American Journal of Public Health 98, no. 6 (2008): 974–85, https://doi.org/10.2105/AJPH.2007.110593.

due to a surplus from World War II. By the 1960s, there was an epidemic of amphetamine dependency and abuse in the United States.[31] They were made illegal for non-prescription use in the early 1970s, classified in the same way as cocaine and morphine, and their use dropped. That is, until the FDA approved amphetamines for the treatment of ADHD in 1996.[32] Drugs marketed for ADHD like Adderall, Vyvanse, et cetera, are amphetamines, and they work quite well to increase alertness and focus. Current exact rates are hard to come by, but it seems that up to 35 percent of college students use stimulants at some point during their undergraduate years.[33] For people without a genuine clinical need for amphetamines, their use can be problematic. They are effective for helping people complete work that relies on skills they already have, but far less effective for learning new skills. They are lousy for contemplative thinking, too, and creative endeavors. And they are very habit forming, and in many cases addictive.

That's the history. And while getting and staying focused has always been hard, it seems it's now harder than ever.

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[31] Rasmussen, "America's First Amphetamine Epidemic 1929–1971."

[32] Florence T. Bourgeois et al., "Premarket Safety and Efficacy Studies for ADHD Medications in Children," PLOS One 9, no. 7 (2014): e102249, https://doi.org/10.1371/journal.pone.0102249.

[33] Amber N. Edinoff et al., "Prescription Stimulants in College and Medical Students: A Narrative Review of Misuse, Cognitive Impact, and Adverse Effects," Psychiatry International 3, no. 3 (July 2022): 221–35, https://doi.org/10.3390/psychiatryint3030018.

The good news is that accessing a state of focus necessary for SDAP is possible for anyone. It starts with understanding the states of mind and body you're after. You'll recall that negative experiences automatically set off the learning process. No coffee, no tutor, no motivational video or mantra required. The fast, massive increase in sympathetic tone caused by adrenaline and norepinephrine ramps up activity in relevant brain circuits. It's proof that if the right electrochemical systems get triggered, we all have the capacity to focus. Again, I'm not saying seek out stressful situations. But this points to adrenaline and norepinephrine as potent pro-focus and pro-plasticity molecules. That is knowledge we can use.

Work in the 1990s from neuroscientists James McGaugh and Larry Cahill at UC Irvine showed that if adrenaline and norepinephrine are elevated using drugs, exposure to cold water, or emotionally activating visual images, the resulting increase in sympathetic tone opens the door for the learning process.[34] Interestingly, the beta-blocker propranolol, which prevents an increase in sympathetic tone, impairs learning and memory of emotional events.[35] Conversely, if

[34] Larry Cahill and James L. McGaugh, "Mechanisms of Emotional Arousal and Lasting Declarative Memory," Trends in Neurosciences 21, no. 7 (1998): 294–99, https://doi.org/10.1016/s0166-2236(97)01214-9.

[35] Sereena Pigeon et al., "Impairing Memory Reconsolidation with Propranolol in Healthy and Clinical Samples: A Meta-Analysis," Journal of Psychiatry and Neuroscience 47, no. 2 (2022): E109–22, https://doi.org/10.1503/jpn.210057.

sympathetic tone is artificially increased before, during, or even immediately after a learning bout, memory is enhanced.[36] And not just for emotional events or information.[37]

Importantly, the relationship between autonomic arousal and learning follows an inverted U-shape curve, a finding first documented in 1908 and reconfirmed since.[38] If autonomic arousal is too low, learning doesn't occur. If autonomic arousal is too high, learning also doesn't occur. Only if autonomic arousal is somewhere on the top of the inverted U-shape curve does learning occur.

Keep this in mind anytime you begin SDAP. The key is to enter a learning bout alert but calm, then to move through it in such a way that your level of autonomic arousal continues

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[36] Keith Tully and Vadim Y. Bolshakov, "Emotional Enhancement of Memory: How Norepinephrine Enables Synaptic Plasticity," Molecular Brain 3, article no. 15 (2010), https://doi.org/10.1186/1756-6606-3-15; Larry Cahill et al., "Enhanced Human Memory Consolidation with Post-Learning Stress: Interaction with the Degree of Arousal at Encoding," Learning & Memory 10, no. 4 (2003): 270–74, https://doi.org/10.1101/lm.62403.

[37] Tom Smeets et al., "True or False? Memory Is Differentially Affected by Stress-Induced Cortisol Elevations and Sympathetic Activity at Consolidation and Retrieval," Psychoneuroendocrinology 33, no. 10 (2008): 1378–86, https://doi.org/10.1016/j.psyneuen.2008.07.009.

[38] "Classics in the History of Psychology," Christopher D. Green, York University, n.d., https://psychclassics.yorku.ca/Yerkes/Law/; Lola Beerendonk et al., "A Disinhibitory Circuit Mechanism Explains a General Principle of Peak Performance During Mid-Level Arousal," Proceedings of the National Academy of Sciences 121, no. 5 (2024): e2312898121, https://doi.org/10.1073/pnas.2312898121.

to increase—but never to the point where you start to feel overly stressed.

The best way I can describe that level of autonomic arousal is if I ask you to imagine the kids' game sometimes called "hot hands." It's where one person places their hands out, palms up, and you put your hands on theirs, palms down. The person with palms up flips their hands over to try and tap the top of your hands before you can pull away. You don't know when they will make their move. You're alert and focused, anticipating something will happen. Learn to recognize that state. That's the ideal level of autonomic arousal for a learning bout. With a bit of practice you'll be able to sense when your autonomic arousal is climbing too high or dropping too low, and keep yourself in that ideal state. In other words, you learn to get better at entering the learning process.

Is Being Focused the Same as Being in a Flow State?

No. Don't confuse this state with a flow state. In flow states we forget the rest of the world (and even ourselves), and we do things as if on autopilot.[39] Flow states are ideal for task performance—executing cognitive and motor skills we have

[39] Jeanne Nakamura and Mihaly Csikszentmihalyi, "The Concept of Flow," in Handbook of Positive Psychology, ed. C. R. Snyder and Shane J. Lopez (Oxford University Press, 2002), 89–105.

already mastered. We don't learn in flow states, we implement what we already learned.

The person who first described flow states—the Hungarian psychologist Mihaly Csikszentmihalyi—accurately described this feeling as when the challenge of a task is matched with your existing skill level, not too easy, nor too hard… just challenging enough to engage you and "an absolute sense of control."[40] That last part—"an absolute sense of control"—means your brain already has the neural circuits in place that it needs to execute the task. And as we discussed above, that means no rewiring… no plasticity is triggered, and no learning occurs.

To trigger plasticity and start learning, you must push past your current abilities. You have to get stumped, miss the mark, and most importantly, you have to feel the agitation of making errors. Recognize that as limbic friction, and seek it out.

Ask any elite performer how they got so good at their craft and they will tell you, "Practice, lots of practice." But there are two kinds of practice. There are the hours upon hours spent drilling what we already know how to do and there is

[40] Mihaly Csikszentmihalyi, Flow: The Psychology of Optimal Experience (Harper & Row, 1990); Roberta Antonini Philippe et al., "Achieving Flow: An Exploratory Investigation of Elite College Athletes and Musicians," Frontiers in Psychology 13 (March 29, 2022): 831508, https://doi.org/10.3389/fpsyg.2022.831508.

the practice of pushing ourselves to learn new skills. The latter is accomplished by making errors through genuine attempts. As the great Twyla Tharp, world-renowned choreographer and dancer, put it, “The more you fail in private, the less you will fail in public.”[41] Twyla understands that elite performance requires genuine attempts—failing and failing again (and again), until the gap between one’s goal and actual performance is reduced to zero.

The Learning State

The experiments that perhaps best illustrate the vital role of focused attention in learning come from the work of neuroscientist Mike Merzenich, PhD, and his lab at University of California, San Francisco.[42] The studies were done in the 1990s using monkeys and later adapted to studies in humans. The monkeys were trained to place their fingers on a rotating cylinder that had both coarse and fine bumps laid out in rows across its surface.[43]

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[41] Twyla Tharp, The Creative Habit: Learn It and Use It for Life (Simon & Schuster, 2006).

[42] Dean V. Buonomano and Michael M. Merzenich, “Cortical Plasticity: From Synapses to Maps,” Annual Review of Neuroscience 21 (1998): 149–86, https://doi.org/10.1146/annurev.neuro.21.1.149.

[43] G. H. Recanzone et al., “Topographic Reorganization of the Hand Representation in Cortical Area 3b of Owl Monkeys Trained in a Frequency-Discrimination Task,” Journal of Neurophysiology 67, no. 5 (1992): 1031–56, https://doi.org/10.1152/jn.1992.67.5.1031; G. H. Rencanzone et al., “Plasticity in the Frequency Representation of Primary Auditory Cortex Following Discrimination Training in Adult Owl Monkeys,” Journal of Neuroscience 13, no. 1 (1993): 87–103,

The monkeys quickly learned that if they detected a difference between the coarse bumps and the fine bumps, they earned a reward. To signal when they noticed that difference, they used their other hand to move a joystick. Over time, the researchers made the task harder by making the difference in the bumps progressively smaller, which required more focus from the monkeys. Not only did the monkeys learn to perform the harder task, but the researchers observed significant changes in the brain areas representing sensations in the fingers that were touching the cylinder. In other words, they observed plasticity associated with that learning. These were adult monkeys, too, showing that plasticity was not just confined to developmental critical periods as had long been believed.

That was interesting but it was just the beginning of the story. In other studies from Merzenich, monkeys had to learn to discriminate between different sound tones. In that case the regions of the brain responsible for hearing changed. Perhaps not surprising, except that the monkeys were experiencing bumps on their fingers at the same time they heard the tones. So why did only the auditory areas of the brain change? Because the monkeys received rewards only

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https://doi.org/10.1523/JNEUROSCI.13-01-00087.1993; Nancy N. Byl et al., “A Primate Genesis Model of Focal Dystonia and Repetitive Strain Injury: I. Learning-Induced Dedifferentiation of the Representation of the Hand in the Primary Somatosensory Cortex in Adult Monkeys,” Neurology 47, no. 2 (1996): 508–20, https://doi.org/10.1212/wnl.47.2.508.

for learning sound tones, so that's what they focused on. Together, these two sets of experiments showed that mere experience doesn't cause neuroplasticity. Learning-associated brain plasticity only occurs in the circuits associated with where our attention is placed.

Since those landmark studies, there have been thousands of experiments exploring the conditions required for learning and plasticity. Consider the often-cited studies from neuroscientist Eleanor Maguire, PhD, and colleagues of London taxi drivers, who must learn to navigate the complex streets of their city by memory (this was prior to the advent of GPS and Google Maps, mind you).[44] They found that, compared to bus drivers, taxi drivers had larger hippocampi, a key brain area for encoding new memories. Then there is the study showing more concentrated patterns of brain

[44] Eleanor A. Maguire et al., "London Taxi Drivers and Bus Drivers: A Structural MRI and Neuropsychological Analysis," Hippocampus 16, no. 12 (2006): 1091–1101, https://doi.org/10.1002/hipo.20233; Alvaro Pascual-Leone and Fernando Torres, "Plasticity of the Sensorimotor Cortex Representation of the Reading Finger in Braille Readers," Brain 116, no. 1 (February 1993): 39–52, https://doi.org/10.1093/brain/116.1.39; Thomas Elbert et al., "Increased Cortical Representation of the Fingers of the Left Hand in String Players," Science 270, no. 5234 (October 1995): 305–07, https://doi.org/10.1126/science.270.5234.305; Friederike Spengler et al., "Learning Transfer and Neuronal Plasticity in Humans Trained in Tactile Discrimination," Neuroscience Letters 232, no. 3 (September 1997): 151–54, https://doi.org/10.1016/S0304-3940(97)00602-2; A. Karni and D. Sagi, "Where Practice Makes Perfect in Texture Discrimination: Evidence for Primary Visual Cortex Plasticity," Proceedings of the National Academy of Sciences 88, no. 11 (June 1991): 4966–70, https://doi.org/10.1073/pnas.88.11.496; Pascual-Leone et al., "The Plastic Human Brain Cortex."

activation among professional violinists for finger movements compared to amateurs.[45] And there are studies showing significant brain circuit changes as people learn to juggle,[46] as well as reversion of those newly formed circuits if people don't practice in the first weeks afterward.[47]

All these studies—the taxi drivers, the violinists, the jugglers—suggest that intense focus is a foundational need in order for SDAP to occur. But suggestions aren't proof. There is, however, one study that really hammers home the relationship between focus and plasticity in humans. In 1993, two researchers from Hebrew University in Israel asked people to practice detecting a slight shift in the angle of a line relative to the angles of a bunch of surrounding lines. The people got better at that task.[48] Not surprising.

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[45] M. Lotze et al., "The Musician's Brain: Functional Imaging of Amateurs and Professionals During Performance and Imagery," NeuroImage 20, no. 3 (2003): 1817–29, https://doi.org/10.1016/j.neuroimage.2003.07.018.

[46] Bogdan Draganski et al., "Changes in Grey Matter Induced by Training," Nature 427 (2004): 311–12, https://doi.org/10.1038/427311a.

[47] Robert J. Zatorre et al., "Plasticity in Gray and White: Neuroimaging Changes in Brain Structure During Learning," Nature Neuroscience 15, no. 4 (March 2012): 528–36, https://doi.org/10.1038/nn.3045; Eran Dayan and Leonardo G. Cohen, "Neuroplasticity Subserving Motor Skill Learning," Neuron 72, no. 3 (November 2011): 443–54, https://doi.org/10.1016/j.neuron.2011.10.008; Yongmin Chang, "Reorganization and Plastic Changes of the Human Brain Associated with Skill Learning and Expertise," Frontiers in Human Neuroscience 8 (2014), https://doi.org/10.3389/fnhum.2014.00035.

[48] M. Ahissar and S. Hochstein, "Attentional Control of Early Perceptual Learning," Proceedings of the National Academy of Sciences 90, no. 12 (1993): 5718–22, https://doi.org/10.1073/pnas.90.12.5718.

Also not surprising was that when participants were asked to detect a slight shift in the angles of all the lines, they got better at that task too. What is surprising is that performance on one task did not improve performance on the other task, despite the fact that what they saw in both tasks was exactly the same. The only difference between the two conditions was what the participants were instructed to pay attention to. The finding leads to a profound insight: that the plasticity they experienced was tied to the act of looking for something and not solely to the success of finding that something. As the authors concluded, "Learning will occur only in those neurons that are activated by the stimulus and that are at the same time relevant for the task. Which neurons are relevant [for neuroplasticity] is dictated by the mechanism or strategy used for task performance (and not merely a stimulus-response correlation)."

These results, along with the earlier studies mentioned here, point to the crucial role of focus in plasticity and learning.

Which brings us to the common saying "fire together, wire together."[49] People often use this phrase to capture the idea that habits form from neurons being activated at nearly the exact same time. The thinking goes that neurons firing simultaneously or one after the other causes them to become wired together, which will lead them to readily fire together

[49] Carla J. Shatz, "The Developing Brain," Scientific American, September 1, 1992, https://www.scientificamerican.com/article/the-developing-brain/.

in the future. But the term “fire together, wire together” is actually unrelated to the kind of learning we are talking about in this chapter—achieving new skills, developing new behavioral patterns, et cetera. This is not a trivial point. This phrase is often attached to the false notion that people can learn when in passive or semi-engaged states. It implies that we can simply expose kids to music to enrich their brain. Not so. It also implies that simply reading or doing something repeatedly is enough to create new brain patterns. Alas, not so. Oh and the “fire together, wire together” phrase is constantly misattributed to psychologist Donald Hebb when in fact it was Carla Shatz, the developmental neurobiologist, who said it first—in the context of developmental brain wiring, not learning.[50] The point is this: Whether young or old, if we want to learn, we need to focus intensely on new information, or in the case of cognitive or motor skills, on our attempt to perform them. The good news is that with the right tools and a little practice, anyone can access focus… and learn.

Tools for Focus

As I mentioned in the introduction to this protocol, we humans have long sought tools to make this endeavor easier. Here’s where science puts us today.

There are three categories of tools to get alert and focused.

[50] Shatz, “Developing Brain.”

- 1 Foundational
- 2 Behavioral
- 3 Pharmacological

The tools offered here can be used for learning bouts, as described above, or even for what I call work sessions—time when you're not necessarily tackling new information or skills, but grappling with some work or physical challenge that simply requires intense focus.

FOUNDATIONAL TOOLS FOR FOCUS

These tools are the bedrock upon which all the other suggestions in this protocol depend. Without them, the rest of the tools in this protocol will be far less effective. Get the foundational tools right, and you'll greatly amplify the potency of all others you opt to use.

Foundational Tool 1: Sleep

In chapter 1 when I said that sleep is the foundation of mental health, physical health, and performance, that included learning of all kinds. When we do not get enough quality sleep, or if our sleep timing is altered by more than an hour or two from our normal schedule, our levels of alertness get harder to regulate and so focusing becomes an even bigger challenge. Thus, aim to get adequate sleep using

the protocols in chapter 1. (See Sleep Protocols 1–6 for a reminder on the basics.)

Foundational Tool 2: NSDR

If you've had less than adequate sleep and you're trying to lock in on a learning bout, consider doing ten, twenty, or, if possible, thirty minutes of non-sleep deep rest (NSDR) at some point prior to starting that learning bout. NSDR, while not a replacement for sleep, is your best bet to restore mental and physical vigor if you didn't get enough sleep. As mentioned in Sleep Protocol 8, NSDR is preferable to naps, because it doesn't create sleep inertia so you won't finish a session feeling groggy, nor does it disrupt your normal sleep schedule. A short nap of thirty minutes or less can also restore mental and physical vigor, but midday napping for more than thirty minutes can really mess up your nighttime sleep. Refer to Sleep Protocol 7 for best practices.

Foundational Tool 3: Stress Control

The other key to an effective learning bout is to make sure you don't show up in a state of excessively high sympathetic tone, or what we commonly call stress. Stressing about something other than what you're trying to learn steals your focus. If you're worrying about your to-do list, it's hard to focus on learning a new song on the guitar. And stressing about what you're trying to learn prevents you from staying

in the alert but calm state you're after. Chapter 3 provides tools you can use "offline"—meaning when you aren't actively engaged in a learning bout—as well as tools you can use during a learning bout, such as the physiological sigh. Both types of tools are effective to help maintain your level of arousal near the peak of the inverted U-shape curve that is ideal for learning.

Foundational Tool 4: Nutrition

If you're even mildly dehydrated, your focus won't be at its best.[51] Drink water and, if you need to, add some electrolytes, which aids the rehydration process. The same goes for your blood sugar levels—make sure yours are relatively stable by not reaching for sugary foods when you know you want to be focused.

Chapter 4 described how to eat to fuel your body for all types of activity, and that includes learning bouts and work sessions. Those protocols offered guidelines, but there are options for how you adapt them for your schedule when it comes to learning. For example, in the first two to three hours of my day, I focus best if I'm fasted. I make sure I'm hydrated, and that I'm caffeinated, too, but unless I wake up especially hungry (a rarity for me), I don't eat yet. If I

[51] Stephen P. J. Goodman et al., "The Effect of Active Hypohydration on Cognitive Function: A Systematic Review and Meta-Analysis," Physiology & Behavior 204 (May 15, 2019): 297–308, https://doi.org/10.1016/j.physbeh.2019.03.008.

already exercised that morning, however, I eat, and then I start work.

You don't want to be distracted by hunger or thirst during a learning bout, nor do you want to be lethargic from an overly large or starch-and-fat-laden meal. Figure out what is best for you. And don't burn valuable mental energy trying to get conditions perfect. That effort can easily use up the self-control available to us, which tends to be finite. If we're mentally exhausted from trying to be perfect, we'll have nothing left for the actual task, a state that psychologists sometimes refer to as ego depletion.[52] Studies show that if people do a lot of decision-making prior to a mental task, they perform less well on the task compared to those who don't.[53] That means no obsessing over your pre-learning nutrition. You'll short-circuit your ability to learn. (See chapter 4 to make sure you're hydrating and eating right.)

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[52] Roy F. Baumeister et al., "Self-Control and Limited Willpower: Current Status of Ego Depletion Theory and Research," Current Opinion in Psychology 60, article no. 101882 (December 2024), https://doi.org/10.1016/j.copsyc.2024.101882.

[53] B. J. Schmeichel et al., "Intellectual Performance and Ego Depletion: Role of the Self in Logical Reasoning and Other Information Processing," Journal of Personality and Social Psychology 85, no. 1 (2003): 33–46, https://doi.org/10.1037/0022-3514.85.1.33; Christoph Lindner et al., "A New Perspective on the Interplay Between Self-Control and Cognitive Performance: Modeling Progressive Depletion Patterns," PLOS One 12, no. 6 (2017): e0180149, https://doi.org/10.1371/journal.pone.0180149.

BEHAVIORAL TOOLS FOR FOCUS

The ability to deliberately set and maintain focus is a superpower. It is a skill that carries over to all sorts of professional, educational, physical health, and relational activities, not just to plasticity and learning. That said, one of the best ways to get better at focusing is to engage in learning bouts.

The act of continually adjusting your state of autonomic arousal so it's not too high and not too low, while engaging with new information or new skills is work. It involves a lot of top-down control as neuroscientists say, which as discussed in chapter 3, is a loose reference to how your prefrontal cortex repeatedly directs and redirects your mental attention. And while no magic pill or ninja move will instantly give you that superpower, there are behavioral tools that can significantly improve your ability to focus. Some assist the transition into a state of focus, and others help you stay there.

Behavioral Tool 1: Meditation

Meditation can take many forms. One shown to improve focus is as follows: Set aside about ten to twenty minutes, sit or lie down, close your eyes, and direct your mental attention to the location between your eyes, or concentrate on your breathing, or both. As you strive to maintain those points of focus, thoughts, sensations, and outside noises inevitably

distract you. Keep redirecting your attention back to the location between your eyes and/or breathing, over and over again, as many times as needed until the allotted time elapses.

From novice to expert, no one is expected to maintain perfect focus for the whole session. It's the redirecting of your attention that is the key component. It's analogous to the final few repetitions of a weight lifting set. It's what drives the nervous system changes that make it easier to focus the next time you meditate, or in any other activity. It's training; it's what is meant by the phrase "meditation is a practice."

A study by researchers at New York University found that meditating in this way for just thirteen minutes per day improved participants' ability to deliberately focus and to regulate their emotions, two strong indications that this behavioral tool can enhance and speed up learning that requires concentration.[54] Shorter meditations of five minutes per day have also been shown to help.[55] You may be thinking this seems like the micro meditation described in

[54] Julia C. Basso et al., "Brief, Daily Meditation Enhances Attention, Memory, Mood, and Emotional Regulation in Non-Experienced Meditators," Behavioural Brain Research 356 (January 1, 2019): 208–20, https://doi.org/10.1016/j.bbr.2018.08.023.

[55] Marius Golubickis et al., "Brief Mindfulness-Based Meditation Enhances the Speed of Learning Following Positive Prediction Errors," Quarterly Journal of Experimental Psychology 77, no. 11 (January 2024): 2312–24, https://doi.org/10.1177/17470218241228859.

Stress Control Protocol 4, or any other meditation… and you would be right. All meditation is perceptual exercise where you intentionally limit what you pay attention to. Remember the Isolator? That was a perceptual filter designed to restrict your visual field—that is, intentionally limit what you pay attention to. Knowing that, it should make sense that what you elect to focus on during a meditation is less important than the act of directing, and redirecting, your attention back to the same place. You can close your eyes and focus on the top of your head, your feet, your lower belly, and your breathing, or you can open your eyes and focus on a point on a wall in front of you (allow yourself to blink as needed, of course).

Studies show improvements in focus stem from all such meditations, because meditation is a perceptual exercise for improving focus.[56] It just so happens that it also helps reduce stress,[57] boost mood,[58] and improve sleep.[59] A recent study even showed that meditation can induce a

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[56] Jonathan M. Lieberman et al., “Neurophysiological Mechanisms of Focused Attention Meditation: A Scoping Systematic Review,” Imaging Neuroscience 3 (2025): IMAG.a.14, https://doi.org/10.1162/IMAG.a.14.

[57] Yi-Yuan Tang et al., “Short-Term Meditation Training Improves Attention and Self-Regulation,” Proceedings of the National Academy of Sciences 104, no. 43 (2007): 17152–56, https://doi.org/10.1073/pnas.0707678104.

[58] Basso et al., “Brief, Daily Meditation Enhances Attention.”

[59] Heather L. Rusch et al., “The Effect of Mindfulness Meditation on Sleep Quality: A Systematic Review and Meta-Analysis of Randomized Controlled Trials,” Annals of the New York Academy of Sciences 1445, no. 1 (2019): 5–16, https://doi.org/10.1111/nyas.13996.

similar glymphatic washout process—that is, the removal of metabolic waste from the brain—as observed during sleep—albeit to a lesser degree.[60] With a year or more of experience, meditation can allow for profound insights into how our thoughts arise (and pass) and for some people it can be transformative for understanding consciousness, but those are not the immediate benefits. Nor should anyone wanting to improve focus shy away from meditation thinking they have to experience those sorts of major insights for it to "work." Meditation done simply as a perceptual exercise is powerful. And the effects show up fast: less procrastination, less impulsivity, and better learning.

If you are very limited on time, here's a simple ten-breath meditation to try immediately upon waking, while still lying in bed: Keep your eyes closed and focus on your breathing for ten slow breaths. If your mind wanders before you reach ten, restart at one, and do that as many times as needed until you reach ten focused breaths. Then get up.

After a few days, you'll notice improvement. Of course, five minutes, as described in Stress Control Protocol 4, and eventually ten, fifteen, or twenty minutes of meditation

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[60] Daniel Goleman and Richard J. Davidson, Altered Traits: Science Reveals How Meditation Changes Your Mind, Brain, and Body (Avery, 2017); Bryce A. Keating et al., "Neurofluid Circulation Changes During a Focused Attention Style of Mindfulness Meditation," Proceedings of the National Academy of Sciences U.S.A. 122, no. 49 (December 2025): e2504961122, https://doi.org/10.1073/pnas.2504961122.

would be even more beneficial. I've heard from many people who do this regularly—meaning three to seven times per week—that they can focus better, and redirect their focus if it wanders. You'll experience the other positive effects of meditation explained above, such as stress reduction and improved sleep too. My lab found that just five minutes a day of meditation provided these benefits.[61]

What time of day or night you meditate is far less important than if you do it regularly. In fact, there is no bad time to meditate as long as you're in a safe location. If your back or knees ache from sitting cross-legged on the floor, remember that you don't have to do that. Make yourself comfortable. Sit in a chair or lie down in bed.

In my experience, it is the rare person who adheres to a daily meditation practice consistently over many years. Most become the "I used to meditate every day, I need to get back to it" people, not unlike the "I used to work out every day, I need to get back to it" people. I believe that happens because we shoot for ten to twenty minutes of meditation every day right out the gate. We make progress but then stop when life gets too busy, and the benefits of meditation disappear slowly so we don't notice until the habit is broken that we are struggling to focus.

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[61] Melis Yilmaz Balban et al., "Brief Structured Respiration Practices Enhance Mood and Reduce Physiological Arousal," Cell Reports Medicine 4, no. 1 (2023): 100895, https://doi.org/10.1016/j.xcrm.2022.100895.

That is why I suggest you set a very low bar, just ten breaths. Do that every morning for two weeks, then shift to doing five minutes of meditation as described in Stress Control Protocol 4. Do that three to seven days per week for a month, then increase the duration to ten minutes, still aiming for three to seven days per week for the next month. Then increase it to fifteen minutes three to seven days per week for the following month, then twenty minutes three to seven days per week… indefinitely. Mark the days you do it on your calendar to keep track of your progress.

Or just stick with five minutes if that feels right. Be the person who sticks to things rather than the person talking about all the healthy stuff they used to do.

For the time investment needed, what meditation provides in terms of ability to focus, along with stress reduction, better mood, and improved sleep is extraordinary.

Behavioral Tool 2: Hypnosis

Hypnosis. I know the word alone might cause some readers to hesitate before continuing. But keep reading. Maybe the word hypnosis makes you picture people squawking like a chicken, unaware their friends are laughing until they emerge from a trance. But that is stage hypnosis. Clinical hypnosis is completely different. I wish it was called directed neuroplasticity instead of hypnosis.

Clinical hypnosis is supported by solid research as a tool to treat chronic pain, and to help with smoking cessation, sleep challenges, anxiety, even gut health issues.[62] Through decades of research, including brain imaging and randomized controlled trials, Stanford University psychiatrist David Spiegel and his lab discovered that clinical hypnosis stimulates specific types of plasticity that allow for better autonomic self-regulation.[63] For example, hypnosis appears to reduce activity in the part of the brain called the dorsal anterior cingulate cortex, indicating diminished self-monitoring. An accompanying increase in connection between the dorsolateral prefrontal cortex and the insula indicates that the brain is doing a better job of regulating the body. These and other findings point to a state

[62] Gary Elkins, "Hypnotherapy for Smoking Cessation, Sleep, Pain, and the Potential of Hypnosis Apps," International Journal of Clinical and Experimental Hypnosis 73, no. 1 (2025): 1–3, https://doi.org/10.1080/00207144.2024.2432843; Ernil Hansen et al., "Editorial: Clinical Hypnosis," Frontiers in Psychology 15 (August 21, 2024): 1464449, https://doi.org/10.3389/fpsyg.2024.1464449; Jenny Rosendahl et al., "Meta-Analytic Evidence on the Efficacy of Hypnosis for Mental and Somatic Health Issues: A 20-Year Perspective," Frontiers in Psychology 14 (2024): 1330238, https://doi.org/10.3389/fpsyg.2023.1330238; Leonard S. Milling et al., "A Meta-Analysis of Hypnotic Interventions for Depression Symptoms: High Hopes for Hypnosis?," American Journal of Clinical Hypnosis 61, no. 3 (2019): 227–43, https://doi.org/10.1080/00029157.2018.1489777; Winfried Häuser, "Gut-Directed Hypnosis and Hypnotherapy for Irritable Bowel Syndrome: A Mini-Review," Frontiers in Psychology 15 (2024): 1389911, https://doi.org/10.3389/fpsyg.2024.1389911.

[63] Heidi Jiang et al., "Brain Activity and Functional Connectivity Associated with Hypnosis," Cerebral Cortex 27, no. 8 (2017): 4083–93, https://doi.org/10.1093/cercor/bhw220.

in which people are simultaneously less self-conscious and yet better at self-regulating.

Hypnosis won't allow you to immediately speak a new language or perform the perfect golf swing, but it is very effective for teaching people how to access otherwise inaccessible brain and bodily states, such that they can more easily re-access those states once hypnosis is over. For example, someone with an intense fear of public speaking can experience being alert but calm while onstage in front of a big audience. With just a few sessions and the occasional brush-up session, people can get very good at this kind of autonomic self-regulation. Hypnosis, in other words, is about learning how to set your state of mind.

If you have trouble with focus, or any of the other issues mentioned above, you may consider hiring a licensed clinical hypnotist to work with in person or online. Alternatively, some years back, Dr. Spiegel developed an app called Reveri that provides this tool to all (I'm an advisor for Reveri). It has you listen to a brief recorded hypnosis script (in Dr. Spiegel's voice) on your phone, a few times per week. Reveri has hypnosis scripts for improving focus, and also for improving sleep, reducing chronic pain, and much more. Each takes about eight minutes to complete. If you try Reveri, remember this: It's not magic. It's also not mind control and it's most definitely not stage hypnosis. It is a tool

for teaching your brain how to access different states, including focus.

Behavioral Tool 3: 40 Hertz Binaural Beats, 20 Hertz Oscillatory Induction, and White, Brown, and Pink Noise

The use of sound as a tool for focus is all about entrainment, or the matching of our internal rhythms to an external sensory event—in these cases, the frequency of sound waves. You may remember the term entrainment from chapter 1 and chapter 5. There, I talked about entraining your circadian rhythm to light, for example, by viewing bright light in the first hour after waking up. There are also ways to entrain to events in the outside world. For example, when you exercise at roughly the same time of day, you experience an increase in catecholamines and certain hormones at that time. Then after about three days or so, you automatically start to feel an increase in energy just prior to when you usually exercise. Entrainment works based on your nervous system's expectation of certain things happening.

Sound waves can entrain our brain, too, and they do so very quickly. When we hear certain frequencies or rhythms, our brain starts to entrain to the predictable features in the sound by our neurons firing in regular fashion. Scientists have figured out how to leverage that and have created specific

sound patterns that increase brain activity in areas that underlie focused states.

One example is binaural beats. Two distinct frequencies of sounds are played in each ear. When that happens, a cluster of neurons in your brainstem called the medial superior olivary nucleus performs a subtraction of the two frequencies. So if a 400 hertz (Hz) sound arrives at one ear and 440 Hz at the other ear, it leads you to perceive a 40 Hz beat. That sound frequency, in particular, has been shown to improve levels of focus, and even certain forms of memory.[64] The precise mechanisms for this are still being parsed but there is evidence that 40 Hz sound can do this in part by causing release of acetylcholine from the medial superior olivary nucleus and the release of dopamine in the striatum, a brain area involved in movement and motivated attention.[65]

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[64] Lorenza S. Colzato et al., “More Attentional Focusing Through Binaural Beats: Evidence from the Global-Local Task,” Psychological Research 81, no. 1 (2017): 271–77, https://doi.org/10.1007/s00426-015-0727-0; Nantawachara Jirakittayakorn and Yodchanan Wongsawat, “Brain Responses to 40-Hz Binaural Beat and Effects on Emotion and Memory,” International Journal of Psychophysiology 120 (October 2017): 96–107, https://doi.org/10.1016/j.ijpsycho.2017.07.010.

[65] Jung M. Park and Li-Huei Tsai, “Innovations in Noninvasive Sensory Stimulation Treatments to Combat Alzheimer’s Disease,” PLOS Biology 23, no. 2 (2025): e3003046, https://doi.org/10.1371/journal.pbio.3003046; Lee Bartel and Abdullah Mosabbir, “Possible Mechanisms for the Effects of Sound Vibration on Human Health,” Healthcare 9, no. 5 (2021): 597, https://doi.org/10.3390/healthcare9050597.

Some evidence even shows that 40 Hz sounds can trigger the brain circuits involved in focus and memory to enter what are called gamma oscillation states, a pattern of neural activity that is favorable for learning.[66] (Interestingly, there is parallel evidence that activating these circuits using clinical devices may also help offset some of the early symptoms of age-related cognitive decline in Alzheimer's disease.[67])

Not everyone appears to benefit from this tool however. It seems if people already have elevated levels of these neuromodulators prior to the sound exposure, the benefits of binaural beats are less apparent or even absent.[68] Alas, without doing an fMRI and blood draws, there is no way to know your baseline dopamine and acetylcholine levels. Considering 40 Hz binaural beats are safe and cost nothing (there are dozens of free apps, videos, and audio recordings of these on Apple music, Spotify, and YouTube), why not try them and see? You have nothing to lose and perhaps something significant to gain in your efforts to focus better.

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[66] Xixi Chen et al., "Gamma Oscillations and Application of 40-Hz Audiovisual Stimulation to Improve Brain Function," Brain and Behavior 12, no. 12 (2022): e2811, https://doi.org/10.1002/brb3.2811.

[67] Artemis Traikapi and Nikos Konstantinou, "Gamma Oscillations in Alzheimer's Disease and Their Potential Therapeutic Role," Frontiers in Systems Neuroscience 15 (December 12, 2021): 782399, https://doi.org/10.3389/fnsys.2021.782399.

[68] Susan A. Reedijk et al., "Eliminating the Attentional Blink Through Binaural Beats: A Case for Tailored Cognitive Enhancement," Frontiers in Psychiatry 6 (June 3, 2015): 82, https://doi.org/10.3389/fpsyt.2015.00082.

Here's how: Try listening to 40 Hz binaural beats for five minutes prior to a learning bout or work session. Once you determine if that works for you, try listening to the beats continuously throughout. (Of course, if you are trying to learn something that requires you to listen to instructions, you'll only be able to use the 40 Hz tones prior to the learning bout.) Also, note that it's recommended you use headphones for binaural beats as it's the best way to entrain your olivary nucleus and other brain circuits to 40 Hz gamma frequencies.[69]

There are exciting new data suggesting that sounds of about 12 to 20 Hz—which cause the brain to entrain to beta, not gamma, waves—may also improve concentration and reduce mental fatigue.[70] Music with 20 Hz as the dominant frequency may be especially effective here.

I personally have tried 40 Hz binaural beats, using one of the many free binaural beat apps out there (I like BrainWave). I've also used white and pink noise, both of which have been shown to improve focus, albeit in a small number of studies.[71] I've heard anecdotal evidence that brown noise

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[69] Colzato et al., "More Attentional Focusing."

[70] Michèle Moessinger et al., "Auditive Beta Stimulation as a Countermeasure Against Driver Fatigue," PLOS One 16, no. 1 (2021): e0245251, https://doi.org/10.1371/journal.pone.0245251;

[71] Othman et al., "Low Intensity White Noise"; Junhong Zhou et al., "Pink Noise: Effect on Complexity Synchronization of Brain Activity and Sleep Consolidation," Journal of Theoretical Biology 306 (August 7, 2012): 68–72, https://doi.org/10.1016/j.jtbi.2012.04.006.

can also be helpful. (In case you're wondering, white noise is noise with all sound frequencies present at equal energy, pink noise sounds less "hissy" than white noise, and brown noise is lower pitched than either white or pink noise.) Brown and pink noise work a bit to increase my focus but I was not terribly impressed by the magnitude of the effect. I have also tried Brain.fm and, for me, it works well (I have no affiliation with Brain.fm, I just like their product). I listen to it while reading, and during learning bouts and even writing sessions lasting anywhere from one to three hours. (I do take a break every fifty to ninety minutes, and avoid looking at my phone in those breaks… more on phone use to come.)

Behavioral Tool 4: Music

The data on listening to music to enhance focus is very clear: Silence is best. (That is unless the music is 12–20 Hz binaural beats, per the above.)

If you like to listen to music when reading or learning, music with no lyrics is better.[72] All that said, some people like to work in cafés, in airports, or in busy offices. The backdrop of auditory noise helps them focus. I suspect there is a lot of

[72] Alessandra S. Souza and Luís Carlos Leal Barbosa, "Should We Turn Off the Music? Music with Lyrics Interferes with Cognitive Tasks," Journal of Cognition 6, no. 1 (2023): 24, https://doi.org/10.5334/joc.273; Lucia De Francesco et al., "What You Listen to Makes a Difference: The Impact of Music on Attention and Well-Being," Annals of the New York Academy of Sciences 1552, no. 1 (2025): 273–83, https://doi.org/10.1111/nyas.70049.

individual variation when it comes to this. You'll have to experiment and see what works for you, and when.

As I write this I am in a completely silent room and it's early evening—the only audible noise is the air conditioner and the keys of my computer, but this morning I listened to Brain.fm for two full hours while I worked on an earlier part of this chapter. I felt extremely focused during both sessions, so go figure. Be a scientist of yourself, and customize your own protocols for focus. One thing is certain: Sound can be a powerful tool for entraining brain states required for learning.

A note before we continue: Throughout this chapter I've been highlighting the important role that adrenaline and norepinephrine play in focus, and the formation of memories and learning. Soon we will cover ways to boost these and some other closely related neuromodulators, like dopamine, using pharmacology. But first, there are established behavioral tools to significantly and safely increase adrenaline and norepinephrine without taking anything. These won't surprise you because I covered them earlier, in different contexts.

Behavioral Tool 5: Physical Exercise

Physical exercise causes meaningful increases in adrenaline through now-well-understood neural circuits, stemming

from your motor cortex in your brain all the way down to your adrenal glands near your lower back.[73] Any vigorous physical activity that involves the large musculature of your legs and core (abdominals, lower back, etc.), will trigger the release of adrenaline, in a manner that scales with the intensity of the activity. Think: compound lower body exercises like squats or deadlifts, running, martial arts, tennis, and on and on. Adrenaline does not cross the blood-brain barrier, but it binds to receptors on your vagus nerve—the two-way neural superhighway that bridges your body and brain.[74] That, in turn, stimulates the release of norepinephrine from your locus coeruleus (remember that, from chapter 1?) and other brain areas involved in generating

[73] Kai Chen et al., "Exercise Training Improves Motor Skill Learning via Selective Activation of mTOR," Science Advances 5, no. 7 (2019): eaaw1888, https://doi.org/10.1126/sciadv.aaw1888; H. G. Laurie Rauch et al., "Neural Correlates of Motor Vigour and Motor Urgency During Exercise," Sports Medicine 43, no. 4 (2013): 227–41, https://doi.org/10.1007/s40279-013-0025-1; Travis Anderson et al., "Exercise and the Hypothalamic-Pituitary-Adrenal Axis: A Special Focus on Acute Cortisol and Growth Hormone Responses," Current Opinion in Endocrine and Metabolic Research 9 (2019): 74–77, https://doi.org/10.1016/j.coemr.2019.08.002; Cristina Mennitti et al., "How Does Physical Activity Modulate Hormone Responses?," Biomolecules 14, no. 11 (2024): 1418, https://doi.org/10.3390/biom14111418; Richard P. Dum et al., "The Mind-Body Problem: Circuits That Link the Cerebral Cortex to the Adrenal Medulla," Proceedings of the National Academy of Sciences 116, no. 51 (2019): 26321–28, https://doi.org/10.1073/pnas.1902297116.

[74] T. Miyashita and C. L. Williams, "Epinephrine Administration Increases Neural Impulses Propagated Along the Vagus Nerve: Role of Peripheral β-Adrenergic Receptors," Neurobiology of Learning and Memory 85, no. 2 (2006): 116–24, https://doi.org/10.1016/j.nlm.2005.08.013.

a state of focus.[75] That's one mechanism, but there are others at play, including increased blood flow and brain oxygenation, and lactate-induced elevations in brain-derived neurotrophic factor (BDNF) (which you'll recall from chapter 2… see how all of this comes together?). The effects of all this on your focus and cognition are significant, and can last several hours afterward.

One of the best ways to use exercise as a tool for focus is brief high-intensity interval training (HIIT)—for example, after a five- to ten-minute warm-up, I like to do a thirty-second near-all-out sprint, followed by thirty seconds of rest, and again, for a total of eight sprint-rest cycles.

Other HIIT options work too. You can find dozens of ideas for these online with a simple web search. It's crucial to warm up for a few minutes before the first round and important too that you select a form of exercise that you can perform without getting injured. I like to run outside on a track or use a stationary assault bike, but you may prefer to swim, use a rower, a stair climber, et cetera.

There are also studies showing improved focus and cognition after sustained twenty to forty-five minute cardio training sessions at lower intensity (60–75 percent of

[75] Dona Lee Wong et al., "Epinephrine: A Short- and Long-Term Regulator of Stress and Development of Illness: A Potential New Role for Epinephrine in Stress," Cellular and Molecular Neurobiology 32, no. 5 (2012): 737–48, https://doi.org/10.1007/s10571-011-9768-0.

maximum heart rate), although the effects of this on focus appear more transient compared to more intense exercise.[76] That makes sense, given HIIT elevates adrenaline and norepinephrine more than lower intensity steady state cardio.[77]

Fewer studies have explored how focus might be increased after resistance training. But assuming the sessions are not overly long or intense, I would expect them to impart a similar focus-enhancing effect as cardio training sessions (see chapter 2 for more on the ideal duration and intensity for those sessions).

Experiment to find the form of exercise you like best to enhance focus. Make sure to time the duration and adjust the intensity of effort so it doesn't exhaust you.

Behavioral Tool 6: Proper Breathing

Sleep apnea is a condition where people under-breathe during sleep. Nasal obstructions, excessive weight, and certain medications cause it. If you snore, there's a good chance you have sleep apnea.

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[76] Kirk I. Erickson et al., "Physical Activity, Cognition, and Brain Outcomes," Medicine & Science in Sports & Exercise 51, no. 6 (June 2019): 1242–51, https://doi.org/10.1249/mss.0000000000001936.

[77] Neil L. Coplan et al., "Exercise-Related Changes in Serum Catecholamines and Potassium: Effect of Sustained Exercise Above and Below Lactate Threshold," American Heart Journal 117, no. 5 (1989): 1070–75, https://doi.org/10.1016/0002-8703(89)90864-8; Mennitti et al., "How Does Physical Activity Modulate Hormone Responses?"

The result? Your brain is deprived of oxygen at night.

Sleep apnea is a dangerous condition, because the health impacts creep up over time—causing fatigue, cognitive impairments, heart issues, and more.[78] Some of these can be reversed with tools to correct breathing while you sleep, but long-lasting impairments can occur.

Why am I discussing sleep apnea in a protocol on focus? Because many people have daytime apnea that impairs their focus too.[79] Daytime apnea is when you hold your breath while working. Chronic mouth breathers experience the same. When your brain doesn't get enough oxygen, you get tired and you can't focus. To feel alert and focus intensely, you need to fix sleep and any daytime apnea.

Luckily, it's easy to do. First, make sure you're breathing through your nose anytime you don't have to speak, eat, or exercise hard. Second, learn to breathe with your diaphragm, also known as belly breathing, in which your belly expands as you inhale. This approach increases the amount of oxygen in the blood and promotes a more relaxed state. You'll be amazed what a difference that will make for your energy levels and focus.

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[78] Anna Abbasi et al., "A Comprehensive Review of Obstructive Sleep Apnea," Sleep Science 14, no. 2 (2021): 142–54, https://doi.org/10.5935/1984-0063.20200056.

[79] Ju-Yeon Jung and Chang-Ki Kang, "Investigation on the Effect of Oral Breathing on Cognitive Activity Using Functional Brain Imaging," Healthcare 9, no. 6 (2021): 645, https://doi.org/10.3390/healthcare9060645.

Check right now. If you're mouth breathing, switch to nasal breathing. Does your belly move out as you inhale and move in as you exhale through your nose? If so, great, you're nasal diaphragmatic breathing. If you're having trouble, imagine inhaling through your nose as if trying to direct the air backward—in other words, think about drawing the air "in and back" through your nose at an angle parallel to the roof of your mouth, as opposed to inhaling "up" your nostrils toward the top of your head. I know that sounds weird, but try it. You'll notice that you immediately shift. And no, you don't have to think about your breathing all day. After two to three breaths done consciously that way, it will carry on reflexively. Nasal diaphragmatic breathing in the daytime also helps you default to the right breathing pattern when you sleep, which can help offset sleep apnea, so it's a twofer.

A few other issues with breathing to address here. If you have a badly deviated septum or challenged nasal breathing brought about by cartilage growth in your nasal passages, you need to fix that. The simplest way to open your nasal passages is to, again, practice nasal diaphragmatic breathing. Your sinuses are not fixed, but over time this effort will help open them up. Extreme cases of nasal passage collapse or other blockage may require surgery. People can often achieve some results from wearing a magnetic nose bridge for a few hours a day and/or at night. They gently pull your nasal passages open and lead to instant improvement in

airway flow. Yes, they look silly. No, you don't have to wear them in public. And it won't make your nostrils or nose bigger. Intake Breathing makes a product that is easy to use. It certainly helped me breathe better at night and when I wore it for a few hours each day, the improvement in breathing, alertness, and focus was immediate, and significant. (Note: I have no affiliation with Intake Breathing.)

Keep in mind that it's 100 percent normal to be able to breathe more easily through one nostril than the other at any one time, and which nostril switches every ninety minutes or so due to a neural circuit mechanism governed by your autonomic nervous system.[80] That's right: The whole right versus left nostril breathing thing practitioners often talk about in yoga classes is supported by modern science, although the reasons for it remain unclear.

Nasal breathing has been shown to improve learning and memory. The reason is simple. It allows for better blood

[80] D. A. Werntz et al., "Alternating Cerebral Hemispheric Activity and the Lateralization of Autonomic Nervous Function," Human Neurobiology 2, no. 1 (1983): 39–43, https://pubmed.ncbi.nlm.nih.gov/6874437/; David Shannahoff-Khalsa, "Lateralized Rhythms of the Central and Autonomic Nervous Systems," International Journal of Psychophysiology 11, no. 3 (1991): 225–51, https://doi.org/10.1016/0167-8760(91)90017-r; Maria Elide Vanutelli et al., "Breathing Right… or Left! The Effects of Unilateral Nostril Breathing on Psychological and Cognitive Wellbeing: A Pilot Study," Brain Sciences 14, no. 4 (2024): 302, https://doi.org/10.3390/brainsci14040302.

flow and oxygenation of the brain, and thus elevated activity of neural circuits involved in learning and memory.[81]

So find out if you snore. Again, it's an indicator that you may have sleep apnea. There are lots of free apps that record you while you sleep and track your snoring patterns. If you snore, address it. Try different devices and exercises. You may need to get fitted with a CPAP machine. Talk to your doctor to explore your options. Eliminating sleep apnea is a big deal for your health, and it will powerfully impact your ability to feel alert and focused.

Behavioral Tool 7: Cyclic Hyperventilation

Cyclic hyperventilation might sound familiar, but there's new context for this powerful tool.

It's quite simple. You hyperventilate. Intentionally. Not a bunch of quick shallow breaths—that's panting! Instead do ten to twenty cycles of a big deep inhale through your nose, using diaphragmatic breathing as described in the section above, and exhale, letting the air passively "fall out" from your mouth. You don't have to heave into the inhale to get the effect you're seeking. Work up to it gradually, and focus on long, diaphragmatic inhales through the nose. Increase

[81] Christina Zelano et al., "Nasal Respiration Entrains Human Limbic Oscillations and Modulates Cognitive Function," Journal of Neuroscience 36, no. 49 (2016): 12448–67, https://doi.org/10.1523/JNEUROSCI.2586-16.2016.

the vigor after a few repetitions. After ten to twenty breaths this way, you will reliably feel much more alert.

As covered in chapter 3, you can spike your levels of adrenaline and norepinephrine and dopamine—the so-called catecholamines—and thereby increase your alertness and focus, by breathing in this specific way. It works by causing increased discard of carbon dioxide, which is called hypocapnia, and increased alkalinity of your blood pH, which is called alkalosis. That in turn triggers fast and significant increases in adrenaline and norepinephrine release in your body and brain.

If you think breathing to shift catecholamine levels is not very potent, think again. This tool works fast and extremely well to increase alertness, so you can focus on whatever task lies before you. It's a causal relationship. In experiments where people hyperventilated through an air mask with artificially elevated CO2, their epinephrine and norepinephrine levels did not increase, nor did they achieve alkalosis.[82] This tells us it's not just the mechanics of deep breathing, it's about increased discard of CO2. By the way, that is why people having a panic attack are told to breathe into a paper bag; it's to encourage them to re-breathe the

[82] Jacek Sein Anand et al., "Hyperventilation with Maintenance of Isocapnia. An 'Old New' Method in Carbon Monoxide Intoxication," PLOS One 12, no. 1 (2017): e0170621, https://doi.org/10.1371/journal.pone.0170621.

CO_2 they expelled, and bring their epinephrine and norepinephrine back to baseline.

When you do cyclic hyperventilation, you won't have a panic attack as long as you pay attention to your autonomic state as you do. If you feel alert after ten breaths, stop. If not, do ten more breaths. If you want, complete twenty breaths, then take a long exhale and hold your breath for ten to sixty seconds until you feel the gasp reflex, and then start another round of cyclic hyperventilation. When you're as alert as you want to be, resume normal nasal diaphragmatic breathing.

Sure, cyclic hyperventilation looks a little wacky but so what? It costs nothing, and it works. It may also have benefits for your immune system.[83] Just don't ever do it in or near water, or while driving, because when your CO_2 is low, the impulse to breathe is delayed. People have drowned practicing this breathing pattern as a way to extend how long they can hold their breath underwater.

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[83] Omar Almahayni and Lucy Hammond, "Does the Wim Hof Method Have a Beneficial Impact on Physiological and Psychological Outcomes in Healthy and Non-Healthy Participants? A Systematic Review," PLOS One 19, no. 3 (March 2024): e0286933, https://doi.org/10.1371/journal.pone.0286933; Sascha Ketelhut et al., "The Effectiveness of the Wim Hof Method on Cardiac Autonomic Function, Blood Pressure, Arterial Compliance, and Different Psychological Parameters," Scientific Reports 13, no. 1 (October 2023): 17517, https://doi.org/10.1038/s41598-023-44902-0.

Behavioral Tool 8: Deliberate Cold Exposure

Deliberate cold exposure works very well to increase alertness and focus for hours afterward. That's not surprising: As you now know, it can more than double or triple your circulating levels of epinephrine and norepinephrine and dopamine—our now-familiar friends, the catecholamines—and it does it within seconds. The data on deliberate cold exposure and learning and memory are mixed because most studies have explored if people can learn during deliberate cold exposure, while they are in the cold.[84] You can already guess what happens there: Deliberate cold exposure generally puts people at the far end of the inverted U-shape curve of autonomic arousal, where focusing ability sharply drops off. That is not the idea here.

This is: Try thirty seconds to three minutes of deliberate cold exposure, then shower and start your learning bout or work session. It's about using deliberate cold exposure to create a long-lasting window of moderate catecholamine increase. Depending on the duration of the cold exposure, the increase in alertness and focus can last for many hours afterward.

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[84] Marika Falla et al., "The Effect of Cold Exposure on Cognitive Performance in Healthy Adults: A Systematic Review," International Journal of Environmental Research and Public Health 18, no. 18 (2021): 9725, https://doi.org/10.3390/ijerph18189725.

The challenge is there is no one-size-fits-all protocol for this one because people's cold tolerance varies.

My suggestion is that if you decide to use deliberate cold exposure as a tool to increase focus, select your method, then choose an uncomfortably cold (for you) but safe temperature. Refer to Stress Control Protocol 2 for more information. If you're using a shower, get all the way under the water. If you're submerging yourself in a plunge or pool, submerge your feet, legs, and torso completely, up to your neck. And yes, put your hands in the water, too, unless you need to grip the sides for safety. The water should be cold, but shouldn't cause pain. Remember that the first ten to thirty seconds are going to feel like mild panic. Your breath will quicken, but then try long, slow, and controlled inhales and exhales. You can opt to distract yourself, or not. It makes no difference. Deliberate cold exposure bypasses most all attempts to reframe the experience as enjoyable. Pretty much the only thing that helps is to remember that it never feels good to be in the cold, but you'll feel great when you get out.

When you do get out, towel off and take a warm or hot shower (or if you're a real masochist, cold), get dressed, and go about your day. Again, to derive the focus-increasing benefits of deliberate cold exposure you don't have to do it immediately before a learning bout or work session because

the effects last hours. Enjoy! (Getting out of the cold, that is.)

PHARMACOLOGICAL TOOLS FOR INCREASING FOCUS

“What can I take to focus?” is one of the most common questions I get. So here goes.

Pharmacological Tool 1: Caffeine

With 90 percent of the world’s adult population drinking caffeine every day, it’s hard to run a study on how caffeine affects alertness and focus… because most participants would be caffeine adapted, meaning they drink caffeine just to achieve normal, or what I call baseline, levels of alertness. This is an important concept to understand.

So many people rely on caffeine for basic alertness that pretty much any experiment comparing one group that takes caffeine to another that doesn’t ends up telling us more about the effects of caffeine withdrawal in the control group than anything else. Alas, most people just won’t quit caffeine for two to three weeks—which is the duration needed to get through the withdrawal—to participate in research. So we are left with very limited studies, all of which point to caffeine as a good way to enhance alertness, but not focus per se.

Let me explain. As we discussed in chapter 1, caffeine staves off fatigue by thwarting the action of adenosine, a molecule that builds up in our brain while we are awake and that causes us to feel sleepy as a result. If you drink caffeine regularly, you're getting some modest focus-enhancing effects, as alertness is a prerequisite for focus.

Caffeine also increases adrenaline and norepinephrine in a dose-dependent manner.[85] A single cup of strong coffee—about 250 milligrams of caffeine—can increase circulating adrenaline by more than 200 percent over baseline levels, and norepinephrine 75 percent over baseline levels.[86] Ingesting caffeine daily also may increase the density of brain dopamine receptors, allowing the same amount of dopamine to impart a greater effect.[87] The largest alertness and focus-enhancing effects of caffeine are likely to be felt if you don't regularly drink caffeine, or if you were to ingest more caffeine than is typical for you.

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[85] T. E. Graham and L. L. Spriet, "Metabolic, Catecholamine, and Exercise Performance Responses to Various Doses of Caffeine," Journal of Applied Physiology 78, no. 3 (1995): 867–74, https://doi.org/10.1152/jappl.1995.78.3.867; Christos Papadelis et al., "Effects of Mental Workload and Caffeine on Catecholamines and Blood Pressure Compared to Performance Variations," Brain and Cognition 51, no. 1 (2003): 143–54, https://doi.org/10.1016/s0278-2626(02)00530-4.

[86] David Robertson et al., "Effects of Caffeine on Plasma Renin Activity, Catecholamines and Blood Pressure," New England Journal of Medicine 298, no. 4 (January 1978): 181–86, https://doi.org/10.1056/nejm197801262980403.

[87] N. D. Volkow et al., "Caffeine Increases Striatal Dopamine D2/D3 Receptor Availability in the Human Brain," Translational Psychiatry 5, no. 4 (2015): e549, https://doi.org/10.1038/tp.2015.46.

A note of caution: Remember the inverted U-shape curve that relates arousal and focus? If you are not a regular caffeine user and you suddenly consume 100 to 250 milligrams more caffeine than is typical for you, it will excessively elevate your sympathetic tone, and you may find it challenging to focus.

I drink caffeine every day, usually in the form of cold brewed or hot yerba maté. I have done that for more than three decades, and by now my caffeine tolerance is very high. Like most people, I swear that it helps me focus though it probably just does that by bringing my levels of alertness up to baseline. If I were to stop drinking caffeine, and push through the withdrawal, I could find my non-caffeinated baseline level of alertness… but I'm uninterested in doing that experiment. I love yerba maté and I love coffee. I sip caffeine from about an hour after I wake up until about 2:00 p.m. daily, and most days I do at least one, and often two or three, long learning bouts or work sessions in that time frame. You'll need to experiment to see what is best for you. I know people who avoid caffeine entirely, those that only use it as a tool for alertness and focus, and I know many like me who use it every day. (Return to Sleep Protocol 11 for more about how caffeine affects your sleep as you experiment with this tool.)

Pharmacological Tool 2: Theanine

I suggest at some point you try taking 100 milligrams of theanine alongside your caffeine. The caffeine-theanine combination can heighten focus, even for those who consume caffeine regularly.[88] What's more, the combination of theanine and caffeine has been shown to improve task-switching capacity and reduce susceptibility to distractions,[89] in part because it's thought to increase alpha wave brain activity, which allows for selective attention.[90] It also increases levels of the inhibitory transmitter GABA, serotonin, and dopamine, which can help make the alert state created by caffeine feel a little more relaxed. For non-regular caffeine users, it reduces any jitteriness caffeine might cause, as it prevents some of caffeine's vasoconstrictive and blood pressure elevating effects.[91] In fact, even though at doses of

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[88] Gayani S. Nawarathna et al., "High-Dose L-Theanine-Caffeine Combination Improves Neurobehavioural and Neurophysiological Measures of Selective Attention in Acutely Sleep-Deprived Young Adults: A Double-Blind, Placebo-Controlled Crossover Study," British Journal of Nutrition 134, no. 3 (2025): 195–204, https://doi.org/10.1017/S0007114525104169.

[89] Suzanne J. L. Einöther et al., "L-Theanine and Caffeine Improve Task Switching but Not Intersensory Attention or Subjective Alertness," Appetite 54, no. 2 (2010): 406–9, https://doi.org/10.1016/j.appet.2010.01.003.

[90] Simon P. Kelly et al., "L-Theanine and Caffeine in Combination Affect Human Cognition as Evidenced by Oscillatory Alpha-Band Activity and Attention Task Performance," Journal of Nutrition 138, no. 8 (2008): 1572S–77S, https://doi.org/10.1093/jn/138.8.1572S.

[91] Ai Yoto et al., "Effects of L-Theanine or Caffeine Intake on Changes in Blood Pressure Under Physical and Psychological Stresses," Journal of Physiological Anthropology 31, no. 1 (2012): 28, https://doi.org/10.1186/1880-6805-31-28.

100 to 400 milligrams (depending on someone's body weight and tolerance) caffeine is considered safe for adults without anxiety, taking 100 to 200 milligrams theanine with it may further improve caffeine's safety profile for non-regular and regular users alike.

Theanine may even allow you to get more of a focus-enhancing effect from less caffeine. In one study, people who ingested as little as 40 milligrams of caffeine combined with 100 milligrams theanine experienced improved focus.[92] Keep in mind that most commercially sold 12 ounce coffees deliver about 200 to 300 milligrams of caffeine depending on the roast, and 16 ounce coffees can contain as much as 400 to 500 milligrams.

Together, the caffeine-theanine combination tends to induce the "alert but calm" state of mind favorable for learning bouts. With its focus-enhancing effect, and various health benefits—such as reduced blood pressure and anxiety—I consider theanine a good tool for caffeine users, which means for nearly everyone.

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[92] T. Giesbrecht et al., "The Combination of L-Theanine and Caffeine Improves Cognitive Performance and Increases Subjective Alertness," Nutritional Neuroscience 13, no. 6 (2010): 283–90, https://doi.org/10.1179/147683010X12611460764840.

Pharmacological Tool 2: Alpha-GPC

I consider alpha-GPC a valuable tool for increasing focus and energy levels, if it's used correctly and very occasionally.

Even small increases in acetylcholine—a neuromodulator—can improve focus.[93] Alpha-GPC does that by increasing the amount of choline that reaches the brain, which in turn indirectly raises acetylcholine levels.[94] Research has found an enhancement of concentration and cognitive flexibility in subjects who take alpha-GPC about thirty minutes prior to a learning bout,[95] and at least one study showed modest improvement in cognition in people with dementia.[96] Other studies, however, show a minimal effect.[97] The variation in outcomes may reflect people's

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[93] Inge Klinkenberg et al., "Acetylcholine and Attention," Behavioural Brain Research 221, no. 2 (2011): 430–42, https://doi.org/10.1016/j.bbr.2010.11.033.

[94] Yasuhisa Tamura et al., "Alpha-Glycerylphosphorylcholine Increases Motivation in Healthy Volunteers: A Single-Blind, Randomized, Placebo-Controlled Human Study," Nutrients 13, no. 6 (2021): 2091, https://doi.org/10.3390/nu13062091.

[95] Chad M. Kerksick, "Acute Alpha-Glycerylphosphorylcholine Supplementation Enhances Cognitive Performance in Healthy Men," Nutrients 16, no. 23 (2024): 4240, https://doi.org/10.3390/nu16234240.

[96] Getu Gamo Sagaro et al., "Activity of Choline Alphoscerate on Adult-Onset Cognitive Dysfunctions: A Systematic Review and Meta-Analysis," Journal of Alzheimer's Disease 92, no. 1 (2023): 59–70, https://doi.org/10.3233/JAD-221189.

[97] Adam G. Parker et al., "The Effects of Alpha-Glycerylphosphorylcholine, Caffeine or Placebo on Markers of Mood, Cognitive Function, Power, Speed, and Agility," Journal of the International Society of Sports Nutrition 12, no. S1 (2015): P41, https://doi.org/10.1186/1550-2783-12-S1-P41.

baseline levels of acetylcholine before they take alpha-GPC, similar to what we discussed earlier with people's baseline levels of dopamine before they listen to 40 Hz binaural beats. If a person's natural levels of a given neuromodulator are already high, adding more may not help. If natural levels of acetylcholine are low to moderate, however, boosting it seems to have a positive effect.[98]

Alpha-GPC is often added to energy drinks, and most nootropic formulas (that is, substances marketed for improving cognition). If you're looking to experiment with ways to increase focus for late afternoon or nighttime work or workouts, and you don't want to disrupt your sleep, you can try 300 to 600 milligrams of alpha-GPC taken thirty minutes or so before you begin. Limit your intake to two to four times per week, because your system will quickly adjust to it.

If you take alpha-GPC, make sure to take 600 milligrams of garlic extract (in capsule form, to avoid the taste and smell of raw garlic) at some point in the same twenty-four hours, to help offset the increase in TMAO that alpha-GPC can cause.[99] TMAO is a metabolite formed by gut bacteria in

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[98] Elise Demeter and Martin Sarter, "Leveraging the Cortical Cholinergic System to Enhance Attention," Neuropharmacology 64 (January 2013): 294–304, https://doi.org/10.1016/j.neuropharm.2012.06.060.

[99] Zeneng Wang et al., "The Nutritional Supplement L-Alpha Glycerylphosphorylcholine Promotes Atherosclerosis," International Journal of

response to high levels of choline and it can pose a cardiovascular risk if elevated too frequently. Garlic contains an antimicrobial chemical called allicin, which may help offset the buildup of TMAO, and garlic extract is very inexpensive.[100]

Some of you may be wondering if you can combine caffeine and alpha-GPC, or even combine caffeine, theanine, and alpha-GPC. Yes, that combination works very well to increase focus and energy levels… but it might be too stimulating for some people, raise blood pressure too high, and increase anxiety, too, so don't go from none to all three. I can take 200 to 300 milligrams caffeine, 600 milligrams alpha-GPC, and 100 milligrams theanine together before a particularly demanding exercise session or learning bout. I probably take that combination once or twice a month. And I still practice the foundational tools described in this chapter: getting great sleep, practicing NSDR, exercising, et cetera. Stimulants and things that increase neuromodulators tend to work best on that solid foundation. I can't emphasize that enough: Great sleep is still the most important tool for focus. Pharmacologic stimulants taken when you are very sleep-deprived sets you up for cardiac issues. Supplements

Molecular Sciences 22, no. 24 (2021): 13477, https://doi.org/10.3390/ijms222413477.

[100] Wei-Kai Wu et al., "Dietary Allicin Reduces Transformation of L-Carnitine to TMAO Through Impact on Gut Microbiota," Journal of Functional Foods 15 (2015): 408–17, https://doi.org/10.1016/j.jff.2015.04.001.

should be used to augment a stable foundation. Not just for safety, but because that is how they work best.

Pharmacological Tool 3: L-tyrosine

This amino acid is used in the body to synthesize dopamine, as well as adrenaline, norepinephrine, and certain thyroid hormones.[101] It comes from foods we eat but, as a nonessential amino acid, our body can also make it from other compounds if needed. That said, consuming more L-tyrosine from foods or supplements leads to a net increase in catecholamine levels.

L-tyrosine is abundant in nuts and seeds, meats, and some cheeses. Parmesan Reggiano has so much tyrosine that you can often see the L-tyrosine crystals on the surface of the cheese block. Yes, it's safe to eat those L-tyrosine crystals, and yes, that is one of the reasons some people like certain cheeses so much—the taste combined with the dopamine increase it causes is highly reinforcing.

L-tyrosine gets converted into L-dopa by the rate-limiting enzyme tyrosine hydroxylase, which means flooding your system with L-tyrosine is not sufficient to increase dopamine—the amount of tyrosine hydroxylase is the

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[101] Toshiharu Nagatsu, "The Catecholamine System in Health and Disease—Relation to Tyrosine 3-Monooxygenase and Other Catecholamine-Synthesizing Enzymes," Proceedings of the Japan Academy, Series B 82, no. 10 (2006): 388–415, https://doi.org/10.2183/pjab.82.388.

bottleneck on dopamine production.[102] Nonetheless, there is some evidence that supplementing with L-tyrosine can improve alertness, focus, and learning under certain conditions.[103] The problem is that most studies used ridiculously high dosages of L-tyrosine to observe these effects—as much as 8 grams or more delivered in capsule or pure powder form. I do not suggest taking that much all at once!

Anecdotally, many people find that supplementing L-tyrosine at doses of just 500 to 2,000 milligrams can help improve alertness and focus, but the published studies indicate L-tyrosine supplements really only help when people are stressed or fatigued.[104]

I have tried 500 to 1,000 milligrams L-tyrosine and I've tried Mucuna pruriens, an over-the-counter naturally occurring form of L-dopa. Both increased my alertness, focus, and energy levels for a few hours but, at least for me, they both led to a significant dip in energy afterward. Some people

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[102] S. Colette Daubner et al., "Tyrosine Hydroxylase and Regulation of Dopamine Synthesis," Archives of Biochemistry and Biophysics 508, no. 1 (2011): 1–12, https://doi.org/10.1016/j.abb.2010.12.017.

[103] Bryant J. Jongkees et al., "Effect of Tyrosine Supplementation on Clinical and Healthy Populations Under Stress or Cognitive Demands—a Review," Journal of Psychiatric Research 70 (November 2015): 50–57, https://doi.org/10.1016/j.jpsychires.2015.08.014.

[104] Simon N. Young, "L-Tyrosine to Alleviate the Effects of Stress?," Journal of Psychiatry & Neuroscience 32, no. 3 (2007): 224, https://pubmed.ncbi.nlm.nih.gov/17476368/.

love L-tyrosine at 500 to 1,000 milligrams before work or exercise sessions, and swear by the focus-enhancing effects. That is why L-tyrosine is found in a lot of energy drinks and pre-workout formulas. I take it from time to time because it happens to be in one of the energy drinks I sometimes consume. So now to energy drinks.

Pharmacological Tool 4: Energy Drinks

More than a third of the world's adult population consumes at least one energy drink per month, and about 9 percent consumes them every day.[105] Studies of adolescents aged thirteen to nineteen have reported that a quarter of those who regularly consume energy drinks have at least one a day, and another quarter have at least two per day.[106] In 2024, the Centers for Disease Control and Prevention noted that 30 to 50 percent of adolescents report some amount of energy drink consumption.[107] These rates are enormous.

To be clear, if you swallow a capsule of 300 to 600 milligrams of alpha-GPC, another capsule of 100 to 200 milligrams theanine, and down both with 100 to 300

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[105] Gema Aonso-Diego et al., "Prevalence of Energy Drink Consumption World-Wide: A Systematic Review and Meta-Analysis," Addiction 119, no. 3 (2024): 438–63, https://doi.org/10.1111/add.16390.

[106] Lyndsey D. Ruiz and Rachel E. Scherr, "Risk of Energy Drink Consumption to Adolescent Health," American Journal of Lifestyle Medicine 13, no. 1 (2018): 22–25, https://doi.org/10.1177/1559827618803069.

[107] "The Buzz on Energy Drinks," School Nutrition, CDC, July 22, 2024, https://www.cdc.gov/school-nutrition/energy-drinks/index.html.

milligrams of caffeine from coffee, tea, or diet soda, you're basically making your own energy drink. But there is an important difference. Many commercially sold energy drinks contain anywhere from 200 to 400 milligrams caffeine, 400 to 900 milligrams alpha-GPC, and 100 to 200 milligrams theanine per serving, and contain one to two servings per can. Many also include L-tyrosine, and more are starting to include huperzine A, an alkaloid that blocks the enzyme that breaks down acetylcholine, leading to overall increases in acetylcholine levels in the brain.[108] Many energy drinks are also now starting to include racetams, a family of synthetic compounds that lead to increased activity of acetylcholine and other neuromodulators in the brain.[109] Now you know why energy drinks are so habit forming: They are carefully engineered neuromodulator-increasing cocktails that taste like candy. They can even cause withdrawal effects if you regularly drink one or more per day, and then try to suddenly stop.

I rarely drink commercially sold energy drinks, preferring instead to get my caffeine from yerba maté or black coffee,

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[108] Zhong Ming Qian and Ya Ke, "Huperzine A: Is It an Effective Disease-Modifying Drug for Alzheimer's Disease?," Frontiers in Aging Neuroscience 6 (August 18, 2014): 216, https://doi.org/10.3389/fnagi.2014.00216.

[109] Ryusei Satoh et al., "Effects of Smart Drugs on Cholinergic System and Non-Neuronal Acetylcholine in the Mouse Hippocampus: Histopathological Approach," Journal of Clinical Medicine 11, no. 12 (2022): 3310, https://doi.org/10.3390/jcm11123310.

and sometimes also taking theanine and/or alpha-GPC in capsule form. If I do have an energy drink, it's usually before a particularly hard workout, as opposed to a learning bout.

I suggest that kids under the age of eighteen avoid energy drinks or pre-workout formulas, which are similar. We have no idea how all that neuromodulator augmentation impacts brain wiring. I hear from people young and old saying they "can't focus," only to learn that they drink one to three energy drinks per day, or a 20 ounce coffee every morning. That's likely 600 milligrams of caffeine just to feel baseline levels of alertness.

Homeostasis is a nonnegotiable rule of biology. If you want to boost focus and energy levels from caffeine or other substances, a daily high dose is not recommended. Put differently: Whatever stimulant you take every day will soon be needed to merely reach your baseline levels of alertness. Better in my opinion to use stimulants as an augmentation, not a requirement.

That said, I am not going to give up my daily caffeine. I enjoy the taste and the feeling far too much. I drink just enough of it to bring about a mild increase in my level of alertness. Coffee has also been shown to benefit longevity by a variety of mechanisms, and yerba maté and green tea both

have additional health benefits from the antioxidants and polyphenols they contain.[110]

Most days I rely on the foundational tools described here: good sleep, nutrition, and hydration. I practice the brief daily meditation I described, and I add caffeine. These generate the alertness and focus I need for learning bouts, work, and exercise.

I include over-the-counter pharmacologic tools like alpha-GPC and theanine here because they do work, and I and many others have derived benefit from them. I also consider them a far safer and less expensive alternative to nicotine and prescription drugs for focus. Which brings me to nicotine.

Pharmacological Tool 5: Nicotine

Ah, nicotine. Humans love the effects of nicotine so much they keep coming up with new delivery methods and every time there is a health hazard associated with one way, they develop another.

This plant alkaloid has been used for centuries to increase focus. It was put into soldiers' kits in the early 1900s because of its known stimulant and stress-reducing

[110] Anna Gawron-Gzella et al., "Yerba Mate—a Long but Current History," Nutrients 13, no. 11 (2021): 3706, https://doi.org/10.3390/nu13113706.

effects.[111] By the 1960s, more than half of adults in the United States and Europe smoked tobacco daily, with many starting upon waking and only stopping at bedtime.[112]

Nicotine, the primary psychoactive ingredient in tobacco, increases alertness and focus, and also heart rate and blood pressure,[113] but it has the unique property among stimulants of inducing relaxation.[114] It makes you feel "alert yet calm"—that state where the autonomic seesaw is tilted ever so slightly toward increased sympathetic tone. It's the perfect state for focus. It also didn't hurt tobacco sales that smoking alongside a stranger or asking for a light was an

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[111] "Life at the Front in Objects," Imperial War Museums, n.d., https://www.iwm.org.uk/history/life-at-the-front-in-14-objects; "Collection: World War I," Stanford Research into the Impact of Tobacco Advertising, n.d., https://tobacco.stanford.edu/cigarettes/war-aviation/world-war-i/.

[112] National Center for Chronic Disease Prevention and Health Promotion (US) Office on Smoking and Health, "Fifty Years of Change 1964–2014," in The Health Consequences of Smoking—50 Years of Progress: A Report of the Surgeon General (Centers for Disease Control and Prevention, 2014), https://www.ncbi.nlm.nih.gov/books/NBK294310/.

[113] William S. Griesar et al., "Nicotine Effects on Alertness and Spatial Attention in Non-Smokers," Nicotine & Tobacco Research 4, no. 2 (2002): 185–94, https://doi.org/10.1080/14622200210123617; Leonie R. Price and Javier Martinez, "Cardiovascular, Carcinogenic and Reproductive Effects of Nicotine Exposure: A Narrative Review of the Scientific Literature," F1000Research 8 (September 4, 2019): 1586, https://doi.org/10.12688/f1000research.20062.2.

[114] Marina R. Picciotto et al., "Effect of Nicotine and Nicotinic Receptors on Anxiety and Depression," NeuroReport 13, no. 9 (2002): 1097–1106, https://doi.org/10.1097/00001756-200207020-00006; Price and Martinez, "Cardiovascular, Carcinogenic and Reproductive Effects"; A. C. Parrott, "Nesbitt's Paradox Resolved? Stress and Arousal Modulation During Cigarette Smoking," Addiction 93, no. 1 (1998): 27–39, https://doi.org/10.1046/j.1360-0443.1998.931274.x.

icebreaker, and that nicotine reduces appetite. Pair all that with advertising that romanticized the act of smoking, and the cost, foul odor, stained teeth, and stagnant breath that smoking causes all went largely overlooked.

Of course, until people finally realized that smoking, dipping, chewing, or snuffing tobacco is dangerous. It significantly increases cancer and stroke risk.[115] And yet, that went largely overlooked for a long time too. Public health announcements started to make these risks clear in the mid-1960s, but it was a rather slow and reluctant campaign until the 2000s, when the tobacco industry started to be heavily taxed and regulated.[116] By 2010, the tobacco industry—formerly one of the most powerful in the world—saw its sales plummet. I was born in 1975 and I can still remember people smoking in airports and on planes, and how every dining table had an ashtray on it. Even the squat racks in the first commercial gym I joined back in 1993 had ashtrays molded into them so you could set your cigarette down while you performed your set! How times have changed.

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[115] Reena S. Shah and John W. Cole, "Smoking and Stroke: The More You Smoke the More You Stroke," Expert Review of Cardiovascular Therapy 8, no. 7 (2010): 917–32, https://doi.org/10.1586/erc.10.56.

[116] John A. Dani and David J. K. Balfour, "Historical and Current Perspective on Tobacco Use and Nicotine Addiction," Trends in Neurosciences 34, no. 7 (2011): 383–92, https://doi.org/10.1016/j.tins.2011.05.001.

Or have they? From 2005 to 2015, as smoking was progressively banned from most every indoor and even some outdoor public spaces, the use of electronic cigarettes, aka vaping, exploded. Vaping is somewhat safer than smoking, but carries its own risks.[117] Nicotine itself doesn't cause cancer, though the contaminants in vape solutions cause lung damage. Also, since nicotine is highly addictive, people struggle to quit vaping just as much as they struggle to quit cigarettes. Health warnings about vaping have curbed the market significantly in recent years, but because no other substance creates that alert-but-calm state like nicotine does, as they always have, people make sure to find a way to get their fix.

Which is why since 2020 the use of oral nicotine products like pouches, gums, and mints has exploded in popularity.[118] Notice the pattern? I get asked about oral nicotine almost daily. "Is it safe? Is it addictive? Is using four 6 milligram pouches worse than vaping?" Some people even use it when

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[117] Juan S. Izquierdo-Condoy et al., "Direct Health Implications of E-Cigarette Use: A Systematic Scoping Review with Evidence Assessment," Frontiers in Public Health 12 (July 29, 2024): 1427752, https://doi.org/10.3389/fpubh.2024.1427752.

[118] Jessica L. Reid et al., "Awareness and Use of Oral Nicotine Pouches Among Youth and Young Adults, 2022–2024: Repeat Cross-Sectional Surveys in Canada, England, the USA and New Zealand," BMJ Public Health 3, no. 2 (2025): e003457, https://doi.org/10.1136/bmjph-2025-003457; Dae-Hee Han et al., "Nicotine Pouch and E-Cigarette Use and Co-Use Among US Youths in 2023 and 2024," JAMA Network Open 8, no. 4 (2025): e256739, https://doi.org/10.1001/jamanetworkopen.2025.6739.

exercising; millions more use it to work. An increasing number of women and teens are using oral nicotine products now too.

So, here is what you need to know about oral nicotine: It absolutely works to sharply increase focus… if you take it occasionally. Meaning no more than two to three times a week, at a dose of 1 to 6 milligrams each time. If you take nicotine daily, you'll soon need that same dose just to reach your former baseline levels of alertness, and without it you'll be in a mild state of withdrawal. You'll feel foggy and more tired than usual. And it is very habit forming. That is why so few people can limit their oral nicotine use to 3 to 6 milligrams once per day. I hear from countless people who started with that amount, loved the effect, then increased to 3 milligrams twice per day, and now use 4 to 6 milligrams or even 8 milligram pouches pretty much all day long.

There's more to discuss as you consider this tool. Nicotine may also protect against Parkinson's and Alzheimer's. You read that right. Nicotine may prevent the neuronal degeneration underlying Parkinson's and Alzheimer's disease.[119] Clinical trials show, however, that once these

[119] Ahmad Alhowail, "Molecular Insights into the Benefits of Nicotine on Memory and Cognition (Review)," Molecular Medicine Reports 23, no. 6 (2021): 398, https://doi.org/10.3892/mmr.2021.12037.

diseases start, nicotine is unlikely to be of benefit.[120] Whether any neuroprotection that nicotine provides is great enough to offset the other health problems nicotine is known to cause is also still unclear. Remember, nicotine itself doesn't cause cancer—it's not mutagenic. But it may accelerate tumor progression.[121] It also raises blood pressure and causes vasoconstriction, which increases heart attack and stroke risk, and because of its vasoconstrictive effects, it has the potential to cause sexual side effects too, especially in men.[122] Experts on oral health are noticing that patients who use nicotine have more gum disease. Experts on skin health observe that it may cause premature aging of skin.[123]

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[120] Chih-Hung Liang et al., "Nicotine Therapy for Parkinson's Disease: A Meta-Analysis of Randomized Controlled Trials," Biomedicines 13, no. 8 (2025): 1814, https://doi.org/10.3390/biomedicines13081814.

[121] Qi Sun and Chunyuan Jin, "Cell Signaling and Epigenetic Regulation of Nicotine-Induced Carcinogenesis," Environmental Pollution 345 (January 2024): 123426, https://doi.org/10.1016/j.envpol.2024.123426; Courtney Schaal and Srikumar P. Chellappan, "Nicotine-Mediated Cell Proliferation and Tumor Progression in Smoking-Related Cancers," Molecular Cancer Research 12, no. 1 (2014): 14–23, https://doi.org/10.1158/1541-7786.MCR-13-0541.

[122] Anna K. Whitehead et al., "Nicotine and Vascular Dysfunction," Acta Physiologica 231, no. 4 (2021): e13631, https://doi.org/10.1111/apha.13631; Christopher B. Harte and Cindy M. Meston, "Acute Effects of Nicotine on Physiological and Subjective Sexual Arousal in Nonsmoking Men: A Randomized, Double-Blind, Placebo-Controlled Trial," Journal of Sexual Medicine 5, no. 1 (2008): 110–21, https://doi.org/10.1111/j.1743-6109.2007.00637.x; Erratum, Harte and Meston, "Acute Effects of Nicotine," Journal of Sexual Medicine 7, no. 11 (2010): 3803, https://doi.org/10.1111/j.1743-6109.2010.02112_3.x.

[123] Laurent Misery, "Nicotine Effects on Skin: Are They Positive or Negative?," Experimental Dermatology 13, no. 11 (2004): 665–70, https://doi.org/10.1111/j.0906-6705.2004.00274.x.

Both are likely as a result of its vasoconstriction effect (i.e., reduced blood flow to the gums and skin). That all makes sense, as reducing the blood supply means less oxygen and nutrients get delivered to an organ or tissue, and fewer waste products of cellular metabolism are removed from it.

Curious about the oral nicotine hype, I tried 4 milligrams nicotine gum and it worked to make me feel alert but calm. And it worked quickly, like flipping a switch. The next day, however, I felt a bit less mentally sharp. I also felt I needed to clear my throat a lot. I spoke to an MD friend who works on neural pathways for speech, and he told me that nicotine binds to both nicotinic and muscarinic acetylcholine receptors—the latter of which are abundant in the smooth muscles of the throat, larynx, and pharynx. He explained that after nicotine wears off, those muscles go into a slight spasm. Sure enough, taking 4 milligrams nicotine again fixed all that right away… until the next day. That experience was enough for me to end the experiment. Fortunately, the throat issue resolved after about three days.

I share this story because it made me understand the appeal—the focus-enhancing effect of nicotine is remarkable—and yet, people should know all the effects, which I'll recap one more time here to underscore them. There's addiction potential, increase in blood pressure and vasoconstriction, and of course how quickly the cravings

begin and many of the positive effects evaporate. I have friends who are otherwise very health conscious, and they need nicotine to do any sort of focused work or exercise. It's become part of the way they just get to baseline. Are they addicted? The classic definition of addiction is continued use despite harm to self or others. One way that I define addiction is something that causes a progressive narrowing of the things that bring you pleasure. According to both definitions, they are not addicted to nicotine (unless you consider the long-term health harms). But I do think they are dependent on it.

As with all things, my motto is this: Do what you want, but know what you're doing. If you want to use nicotine, use the lowest possible dose that is effective for you. Stick to pouches, mints, or gums because the cancer risk appears to be lower than with cigarettes, but be aware that we do not yet have enough data to know this risk conclusively.[124] Limit your use to two to four times per week, and at most 6 milligrams per use (2–4 milligrams would be even better). That is, if you want the potent focus-enhancing effect of nicotine to last.

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[124] Paulina Natalia Kopa-Stojak and Rafał Pawliczak, "Comparison of Carcinogenic Potential of Alternative Tobacco Products. A Systematic Review," Toxicology Mechanisms and Methods 35, no. 9 (July 2025): 1161–75, https://doi.org/10.1080/15376516.2025.2536664.

Pharmacological Tool 6: Prescription Drugs: Adderall, Ritalin, Modafinil and Armodafinil, Sunosi, Wellbutrin

These are the heavy guns for pharmacologically increasing alertness and focus. Collectively, these drugs are referred to as sympathomimetics: They increase levels of sympathetic tone by directly increasing levels of dopamine and norepinephrine in the brain, albeit each through different mechanisms.[125] Some block reuptake of dopamine and norepinephrine at synapses, leading to more availability of these two neuromodulators in the brain; others stimulate more release and block reuptake. Some also weakly increase other neuromodulators like serotonin. But they all potently increase alertness and focus.

I hesitated to include these drugs in this book, because they were developed to treat specific clinical conditions like ADHD (Adderall, Vyvanse, and Ritalin), or excessive daytime sleepiness (modafinil and Sunosi), or depression and smoking cessation (Wellbutrin), and they were not

[125] Stephen V. Faraone, "The Pharmacology of Amphetamine and Methylphenidate: Relevance to the Neurobiology of Attention-Deficit / Hyperactivity Disorder and Other Psychiatric Comorbidities," Neuroscience & Biobehavioral Reviews 87 (April 2018): 255–70, https://doi.org/10.1016/j.neubiorev.2018.02.001; Mehmet Sofuoglu and R. Andrew Sewell, "Norepinephrine and Stimulant Addiction," Addiction Biology 14, no. 2 (2009): 119–29, https://doi.org/10.1111/j.1369-1600.2008.00138.x; Benjamin P. Kay et al., "Stimulant Medications Affect Arousal and Reward, Not Attention Networks," Cell 188, no. 26 (2025): 7529–46.e20, https://doi.org/10.1016/j.cell.2025.11.039.

developed or marketed for merely increasing alertness and focus. The reality, however, is many millions of people get them prescribed or obtain them illicitly for that purpose. Knowing how they work and their margins of safety therefore feels warranted. Also, I hear often from parents wondering if they should medicate their child for ADHD. Or I hear from someone who thinks they themselves may have mild to severe ADHD, and they want to know what to do about it. They know I am not an MD, but they tell me it is hard to find a US doctor who doesn't almost immediately default to prescribing these medications. They are confused and conflicted about what to do.

Here is what everyone should know: All prescription drugs for focus have side effects. These vary by drug, dosage, and by person in ways that can only be determined empirically, meaning by trial and error. Also—and this is very important—the dosage at which some of these drugs are effective is enormously variable. Adderall works to treat ADHD in some people at 5 milligrams, whereas others need 40 milligrams or even more.[126] It's simply impossible to know how someone will respond, and it requires time to find the minimum effective dose. Of course, all the drugs in this section require a prescription to acquire legally. Illegal and questionably legal sources carry significant risk.

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[126] Preeti Patel et al., "Dextroamphetamine-Amphetamine," StatPearls, last updated April 6, 2025, https://www.ncbi.nlm.nih.gov/books/NBK507808/.

Fentanyl—a deadly opioid—has frequently been found in counterfeit Adderall.[127]

The reason I call these "the heavy guns" is they work so well that people quickly become dependent on them, and some, such as the amphetamine-based stimulants like Adderall, carry abuse potential because they can create euphoria.[128] It's also important to note, however, that failure to treat kids or adults who have ADHD puts them at a much higher risk for addiction than not, due to the impulsivity associated with ADHD.[129] That said, amphetamines may stunt growth,[130] cause sleep disturbances,[131] and even lead to psychosis if abused or taken by people with bipolar depression or a predisposition to psychosis.[132] And all these drugs are hard

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[127] A. Abdul-Sattar et al., "Adderall Laced with Fentanyl: The Epidemic Is Killing Our Youth," Internet Journal of Public Health 9, no. 1 (2023), https://ispub.com/doi/10.5580/IJPH.56911.

[128] Wayne C. Drevets et al., "Amphetamine-Induced Dopamine Release in Human Ventral Striatum Correlates with Euphoria," Biological Psychiatry 49, no. 2 (2001): 81–96, https://doi.org/10.1016/s0006-3223(00)01038-6.

[129] Bahadar S. Srichawla et al., "Attention Deficit Hyperactivity Disorder and Substance Use Disorder: A Narrative Review," Cureus 14, no. 4 (2022): e24068, https://doi.org/10.7759/cureus.24068.

[130] Erin Richardson et al., "Growth Perturbations from Stimulant Medications and Inhaled Corticosteroids," Translational Pediatrics 6, no. 4 (2017): 237–47, https://doi.org/10.21037/tp.2017.09.14.

[131] J. A. Santisteban et al., "Effect of Extended-Release Dexmethylphenidate and Mixed Amphetamine Salts on Sleep: A Double-Blind, Randomized, Crossover Study in Youth with Attention-Deficit Hyperactivity Disorder," CNS Drugs 28, no. 9 (2014): 825–33, https://doi.org/10.1007/s40263-014-0181-3.

[132] Lauren V. Moran et al., "Risk of Incident Psychosis and Mania with Prescription Amphetamines," American Journal of Psychiatry 181, no. 10 (2024): 901–9, https://doi.org/10.1176/appi.ajp.20230329.

to quit.[133] When someone does try, they often suffer from excessive daytime sleepiness and mood and focus issues, aka withdrawal. It's much worse than caffeine or nicotine withdrawal, which is why most people don't quit—they use these drugs indefinitely. Thus, the decision to medicate a child or adult with Adderall or Vyvanse (both amphetamines) or Ritalin (a non-amphetamine stimulant) needs to be carefully considered.

There are also a lot of nondrug therapies for ADHD that can be effective and/or can help reduce dosage needs. They are the foundational and behavioral tools mentioned above. What about supplements? Can alpha-GPC replace a genuine need for Adderall to treat ADHD? Unlikely, but a lifestyle that follows the foundational behaviors described in the protocols here (sleep, hydration, nutrition that favors stable blood sugar levels and satiety, etc.) along with regular meditation or hypnosis for increased focus, and timed use of caffeine and theanine, may allow people to use a lower dose of a prescription drug for ADHD. There are also many cases where these lifestyle changes have allowed people with ADHD to stop taking their medications and still thrive, or thrive even more. It's all highly individual.

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[133] L. L. Weyandt et al., "Prescription Stimulant Medication Misuse: Where Are We and Where Do We Go from Here?," Experimental and Clinical Psychopharmacology 24, no. 5 (2016): 400–414, https://doi.org/10.1037/pha0000093.

That's ADHD. What about the person who doesn't have ADHD, but wants to use a prescription medication to induce states of intense focus one to three times per week? They may be doing everything else correctly, adhering to the foundational tools I've described in this chapter, but they want an "edge." And what about the many millions of people who, due to smartphones, have what clinicians are now calling mild ADHD, which is not a diagnosis but a catch-all for no longer being able to focus? Should they take a sympathomimetic? We can no longer pretend that smartphone-induced mild ADHD doesn't exist. My stance is that we should be aware of all the tools available for focus that are legal, and consider their relative potency and risks.

RITALIN

This non-amphetamine stimulant has largely fallen out of favor for the treatment of ADHD and for focus purposes because it's short acting, and people prefer the stronger, longer lasting effects they get with Adderall and Vyvanse.[134] A recent brain imaging study found that methylphenidate—the active ingredient in Ritalin—did not actually increase activation of attentional networks in the

[134] Dusan Kolar et al., "Treatment of Adults with Attention-Deficit / Hyperactivity Disorder," Neuropsychiatric Disease and Treatment 4, no. 2 (2008): 389–403, https://doi.org/10.2147/ndt.s6985.

brain.[135] It merely reduced the feeling of fatigue and the effects of sleep deprivation on attention. It also made boring work feel more tolerable, and even rewarding—thus allowing for more sustained effort at tasks. The study didn't look at other stimulants, but this may be a general theme of the drugs covered in this section because few if any stimulants directly increase focus. Raising dopamine levels can reduce background "noise" in the brain, and therefore the tendency to get distracted, and raising norepinephrine increases alertness.[136] Together, that can make trivial things feel more engaging. It's important to know that people with ADHD can focus quite well—they just have a much harder time focusing on things they don't enjoy than do people without ADHD. Alas, much of life requires focusing on things we may prefer not doing. Apparently, stimulants like Ritalin trick the brain into ignoring our preferences.

ADDERALL AND VYVANSE

Make no mistake. These are amphetamines, aka speed, but they are not meth. There is a prescription drug for ADHD called Desoxyn, which is pharmaceutical grade methamphetamine, but it's rarely prescribed due to the

[135] Benjamin P. Kay et al., "Stimulant Medications Affect Arousal and Reward, Not Attention Networks," Cell 188, no. 26 (December 2025): 7529–46, https://doi.org/10.1016/j.cell.2025.11.039.

[136] Stephen M. Stahl, "Norepinephrine and Dopamine Regulate Signals and Noise in the Prefrontal Cortex," Journal of Clinical Psychiatry 70, no. 5 (2009): 617–18, https://doi.org/10.4088/jcp.09bs05143.

addiction and abuse potential.[137] (Only a few hundred prescriptions exist in the US currently.)

Adderall and Vyvanse are similar to each other but the former works faster, within minutes, and lasts four to six hours; whereas Vyvanse's effects come on more slowly and last ten or more hours. An estimated 16.5 million people (adults and kids combined) take these amphetamines every day in the United States.[138] Some for ADHD, and some just to help with focus. They get them with valid prescriptions from board-certified MDs. If that is surprising, consider that despite being 5 percent of the world's population, the US accounts for more than 80 percent of the prescribed Adderall and Vyvanse taken worldwide.[139] You can be sure that it is not all being prescribed for clinical levels of ADHD.

Here is my suggestion: If you are considering using these drugs for focus and don't have diagnosed ADHD (and even if you do), first do an inventory of the foundational and behavioral tools for focus that are listed in this chapter. Are

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[137] Anna Moszczynska and Sean Patrick Callan, "Molecular, Behavioral, and Physiological Consequences of Methamphetamine Neurotoxicity: Implications for Treatment," Journal of Pharmacology and Experimental Therapeutics 362, no. 3 (2017): 474–88, https://doi.org/10.1124/jpet.116.238501.

[138] IQVIA, "Stimulant Prescription Trends in the United States from 2012–2023," November 13, 2024, https://www.deadiversion.usdoj.gov/pubs/docs/IQVIA-Report-on-Stimulant-Trends-2024.pdf.

[139] Brian J. Piper et al., "Trends in Use of Prescription Stimulants in the United States and Territories, 2006 to 2016," PLOS One 13, no. 11 (2018): e0206100, https://doi.org/10.1371/journal.pone.0206100.

you getting enough sleep and exercise? Are you drinking enough water? Have you tried brief meditation or hypnosis for focus, and perhaps the caffeine, theanine, and/or alpha-GPC combination I described? If not, I suggest you do before resorting to use of prescription-level amphetamines. (Also, see the section “A Note on Phone Management” at the end of this protocol, as the phone is a severe impediment to focusing for many people.)

To be clear, I am not against prescription drugs per se, and if you do take one, don’t stop suddenly. Talk to your doctor. Many prescription drugs have their place in treatment, but unless there is a genuine clinical need, using stimulants as powerful as Adderall or Vyvanse on a regular basis will soon cause dependence, and other tools you try will pale in comparison.

MODAFINIL AND ARMODAFINIL

These drugs were designed to treat “excessive daytime sleepiness” caused by narcolepsy or sleep apnea or jet lag. Today, they are commonly prescribed to people simply wishing to improve their focus for work or school. They are not amphetamines, but they can similarly disrupt sleep and appetite, and they still pose some cardiac risk—a feature of all sympathomimetics because they elevate blood pressure

and heart rate.[140] These drugs are interesting because while they raise alertness and focus, they also have the rare feature of also improving cognition, in particular executive functions that allow for orchestrated task switching and working memory—the ability to hold relevant information in mind for only as long as is necessary.[141] At the same time, they may impair creative divergent thinking.[142]

In speaking with psychiatrists who prescribe these drugs, I've learned that people's sensitivity to modafinil and armodafinil varies tremendously. Some find 50 milligrams much too stimulating and their sleep is disrupted; others need 200 milligrams to achieve the benefits and sleep just fine. I have tried both armodafinil and modafinil and found they can erase my jet lag but they impair my recall memory. Armodafinil is essentially longer lasting modafinil, but even

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[140] Thomas Roth et al., "Evaluation of the Safety of Modafinil for Treatment of Excessive Sleepiness," Journal of Clinical Sleep Medicine 3, no. 6 (2007): 595–602, https://pubmed.ncbi.nlm.nih.gov/17993041/.

[141] Reyna Lamas-Aguilar et al., "Armodafinil as a Potential Pharmacological Treatment for Attention Deficit Hyperactivity Disorder in Adults: A Review," Current Neuropharmacology 22, no. 11 (2024): 1899–908, https://doi.org/10.2174/1570159X22666240131121642; Erratum, Lamas-Aguilar et al., "Armodafinil as a Potential Pharmacological Treatment," Current Neuropharmacology 23, no. 14 (2025): 2004, https://doi.org/10.2174/1570159X23999250210110914; R. M. Battleday and A.-K. Brem, "Modafinil for Cognitive Neuroenhancement in Healthy Non-Sleep-Deprived Subjects: A Systematic Review," European Neuropsychopharmacology 25, no. 11 (2015): 1865–81, https://doi.org/10.1016/j.euroneuro.2015.07.028.

[142] Ahmed Dahir Mohamed, "The Effects of Modafinil on Convergent and Divergent Thinking of Creativity: A Randomized Controlled Trial," Journal of Creative Behavior 50, no. 4 (2016): 252–67, https://doi.org/10.1002/jocb.73.

at the lowest dose of modafinil (I broke a 100 mg tablet in half), I struggled to fall asleep later that night. That said, I have a colleague who takes 100 to 200 milligrams of modafinil four to five times a week, which was prescribed to him for "mild ADHD," and he has no issues sleeping afterward. These drugs used to be expensive—but the release of generic versions made it less than one dollar per 100 milligram pill. That, combined with their mild cognitive-enhancing effects, makes me think that modafinil and armodafinil use will increase in years to come.

SUNOSI

This is a relatively new drug that also acts to increase dopamine and norepinephrine in the brain.[143] It is FDA approved for offsetting excessive daytime sleepiness due to sleep apnea, and recently it performed well in clinical trials for ADHD,[144] making it likely to soon be approved for that purpose as well. I know people have already started using Sunosi by off-label legal prescription either daily for

[143] Vivien C. Abad, "Profile of Solriamfetol in the Management of Excessive Daytime Sleepiness Associated with Narcolepsy or Obstructive Sleep Apnea: Focus on Patient Selection and Perspectives," Nature and Science of Sleep 13 (January 2021): 75–91, https://doi.org/10.2147/NSS.S245020.

[144] Megan C. Cuomo et al., "Solriamfetol for the Management of Excessive Daytime Sleepiness," Journal of Pharmacy Practice 35, no. 6 (2022): 963–70, https://doi.org/10.1177/08971900211009080; Sheridan M. Hoy, "Solriamfetol: A Review in Excessive Daytime Sleepiness Associated with Narcolepsy and Obstructive Sleep Apnoea," CNS Drugs 37, no. 11 (2023): 1009–20, https://doi.org/10.1007/s40263-023-01040-5.

ADHD[145] or occasionally for enhancing focus. Sunosi is not an amphetamine, and thus has less abuse or addictive potential than the other compounds mentioned above. It does not cause euphoria, nor does it seem to cause withdrawal on days it is not taken. Provided the dosing is correct (typically 75–150 mg taken in the morning), it does not seem to disrupt sleep either. Interestingly, Sunosi is a much lower affinity dopamine and epinephrine reuptake inhibitor than Wellbutrin, Adderall, or modafinil and it may have some mild serotonin-increasing effects too. That might explain why people find it less stimulatory and anxiety promoting than other pro-alertness and focus drugs. As with all drugs that increase dopamine, people with conditions that cause mania or psychosis should be cautious about using Sunosi.

WELLBUTRIN

This dopamine and norepinephrine reuptake inhibitor is much more noradrenergic than dopaminergic as compared to the other compounds in this section.[146] This means it raises norepinephrine more than it raises dopamine. It was developed to help with smoking cessation, and to treat depression, while bypassing the sexual side effects caused by

[145] Ruta Nonacs, "Solriamfetol (Sunosi): Another Option for the Treatment of Adult ADHD?," MGH Psychiatry News, November 17, 2023, https://m7ghpsychnews.org/solriamfetol-sunosi-another-option-for-adhd/.

[146] Maurizio Fava et al., "15 Years of Clinical Experience with Bupropion HCl: From Bupropion to Bupropion SR to Bupropion XL," Primary Care Companion to The Journal of Clinical Psychiatry 7, no. 3 (2005): 106–13,

many SSRIs. Eventually it was adopted by the ADHD community—a mix of psychiatrists and online patient groups that found it useful for its off-label use to increase focus.[147] At higher dosages—200 milligrams or more—Wellbutrin increases risk of seizures in those who are seizure prone, and it carries the usual anxiety and cardiac health warnings of all sympathomimetics.[148] Because of its low dopamine-promoting activity, it has low abuse risk.

I know many people who take 100 to 150 milligrams of Wellbutrin daily—typically the sustained release versions—purely for its energy and focus increasing effects. I have taken 75 to 150 milligrams specifically for that purpose and it worked well, but I found that Wellbutrin impeded my memory, especially for names. I am not sure why or how, but it was a potent and reproducible effect for me. Fortunately, my memory reverted to normal as soon as

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https://doi.org/10.4088/pcc.v07n0305; Barrett R. Cooper et al., "Evidence That the Acute Behavioral and Electrophysiological Effects of Bupropion (Wellbutrin®) Are Mediated by a Noradrenergic Mechanism," Neuropsychopharmacology 11 (1994): 133–41, https://doi.org/10.1038/npp.1994.43; Jianming Dong and Pierre Blier, "Modification of Norepinephrine and Serotonin, but Not Dopamine, Neuron Firing by Sustained Bupropion Treatment," Psychopharmacology 155 (March 2001): 52–57, https://doi.org/10.1007/s002130000665.

147 Wim Verbeeck et al., "Bupropion for Attention Deficit Hyperactivity Disorder (ADHD) in Adults," Cochrane Database of Systematic Reviews 10, no. 10 (2017): CD009504, https://doi.org/10.1002/14651858.CD009504.pub2.

148 "Wellbutrin (Bupropion Hydrochloride) Tablets" (label), Drugs@FDA: FDA-Approved Drugs, revised May 2017, https://www.accessdata.fda.gov/drugsatfda_docs/label/2017/018644s052lbl.pdf.

the drug wore off. I mention this side effect knowing that everyone responds differently to compounds and simply to illustrate how individual the effects of sympathomimetics can be.

A Note on Phone Management

It's clear we messed up. Smartphones are a great technology for many things: texting, apps, and more. And I do think social media can be used to inform, learn, and teach, and of course for entertainment and social connection (of sorts) too. With most powerful technologies, however, humans tend to go overboard before they course-correct. The time for that course correction is now.

People can't seem to set their phones down and even if they do, their phones still interfere with the ability to think. A study that looked at the effect of having your phone on your desk versus in a bag in the same room versus in an entirely separate room found that having your phone on your desk or even zipped up in a backpack reduces your working memory and fluid intelligence—the ability to figure things out on the fly—compared to when your phone is placed in a different room.[149] The authors of the study were able to disentangle focus—people's ability to direct their attention to a task—from how much mental energy they could devote to what they were focusing on. They found that, even when

[149] Skowronek et al., "Mere Presence of a Smartphone."

people could focus on the task at hand, having their phone within physical reach reduced their cognitive ability. Apparently, even if we are not consciously thinking about our phone or seeing notifications, if it's within our physical proximity our brain devotes mental energy to resist the temptation to pick it up. The takeaway thus is simple: To maximize focus, put your smartphone in an entirely different room from where you are.

Another very useful tool is to restrict your social media apps to an old phone that has a completely different phone number. I did this, and now I purposefully limit my time with my "social media phone" to sixty minutes or less per day.

If you do both those things and still find yourself continually looking at your phone, I suggest getting a timed lockbox that has a no-code-out option—meaning there is no way whatsoever to open the box until the timer has elapsed. I use a no-code-out lockbox for my phone, and find it helps for getting work done. I don't have ADHD but I find that once the phone is truly inaccessible, my ability to focus is massively increased.

The best analogy I can come up with for the smartphone as it relates to the topic is this: Just as sleep apnea or daytime apnea reduces the amount of oxygen reaching your brain, smartphones cause attentional and cognitive apnea. They

lessen the amount of neural activity in our brain circuits for focus. To focus on a work session or a learning bout of any kind, get your phone far out of reach. Your mild-to-severe ADHD may be more phone-proximity-driven than you realize. Perhaps partially, perhaps entirely. There is only one way to find out!



LEARNING PROTOCOL

Use Mental Contrasting to Achieve a "State of Urgency" and Reach Your Goals

WHAT TO DO

- Think of being in "a state of urgency," rather than a state of motivation. Urgency is an experience we are all familiar with. We tend to know what it feels like in our mind and body. Motivation can be vague and ephemeral.
- The most reliable method to create this state is a psychological tool called mental contrasting. It asks you to imagine your goals, and the obstacles to achieving them, in a structured way. Even if you're already on your way toward your goals, this strategy can further assist you.

HOW IT WORKS

“Motivation is everything.” We hear this all the time and know it’s true, yet people vary in their abilities to get and stay motivated. Especially to do things that require sustained focus and energy.

The tools I mentioned in the previous protocol were based on physiology as the primary lever. The idea is if we increase sympathetic tone and deploy catecholamines, we can create the neurochemical milieu required for plasticity. I made no mention of psychological tools.

Mental contrasting is a psychological tool, and a powerful one.

Imagine being able to instantly switch on a state of urgency whenever you want. Someday we’ll have that. I’m certain because my colleague at Stanford, neurologist Joe Parvizi, MD, PhD, conducted incredible experiments showing that if activity in a brain area called the anterior midcingulate cortex (aMCC) is increased, people feel an immediate surge in mental energy and readiness to lean into challenges.[150] Alas, currently the only way to instantly and selectively switch on your aMCC is to drill a hole in your skull, lower an electrode into this region, and pass an electrical current

[150] Josef Parvizi et al., “The Will to Persevere Induced by Electrical Stimulation of the Human Cingulate Gyrus,” Neuron 80, no. 6 (2013): 1359–67, https://doi.org/10.1016/j.neuron.2013.10.057.

into it. It will be a while—probably five to seven years—before selective aMCC stimulation can be achieved noninvasively. In the meantime, if we want to activate a state of urgency we're going to have to rely on something else. There is a pill to make you feel more alert and focused, but there is no pill that can make you feel motivated to do a specific task.

Fortunately, a solution does exist. It's this psychological tool that can reliably shift your mindset, allowing you to become more mentally and physically able to exert effort. But first we should get clear on what a mindset is.

People use the word mindset all the time nowadays but in the context of motivation psychology, it has a very specific meaning and purpose. Alia Crum, a professor of psychology and researcher who directs her own lab at Stanford, explained it best: "A mindset is a setting for your brain."[151] Meaning, a mindset is a psychological stance that makes certain thoughts and actions easier to access, and other thoughts and actions harder to access. Alia's definition also speaks to the power of mindsets to simultaneously influence different aspects of your physiology—your brain directs many systems, including your nervous system, your

[151] Alia Crum, "#1 Mindset Expert: Simple Mindset Shifts That Transform Your Body, Energy, & Life," interview by Mel Robbins, December 20, 2025, YouTube, https://www.youtube.com/watch?v=oJAV0rtSrFc.

hormones, even your immune system, in specific ways.[152] That, in turn, guides our behavior. When we switch to a mindset that embraces a challenge, it activates neural circuitries and chemical reactions that make us feel more capable of running that extra mile, putting in extra study hours, or going for one more hour of piano practice. Physiology is also a lever here, but a secondary one.

You may have heard the term "growth mindset," coined by Stanford psychologist Carol Dweck. The term refers to the belief that ability is not fixed but rather can improve with effort.[153] With a growth mindset, a teenager who is failing calculus understands that studying hard will lead to better comprehension. That state unlocks the possibility for even more: You believe in the effort that learning requires, and take the action that it requires too. Upholding a growth mindset means you actively avoid identity-based and valuation labels of being good or bad at something. You aren't "good" or "bad"; you simply haven't put in the effort and taken the action yet. A growth mindset never says, "I'm not a math person"; it says, "I can get better at math if I work at it." Experiments from Carol and others show that adopting

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[152] Alia J. Crum and Ellen J. Langer, "Mind-Set Matters: Exercise and the Placebo Effect," Psychological Science 18, no. 2 (2007): 165–71, https://doi.org/10.1111/j.1467-9280.2007.01867.x.

[153] Carol S. Dweck and David S. Yeager, "Mindsets: A View from Two Eras," Perspectives on Psychological Science 14, no. 3 (2019): 481–96, https://doi.org/10.1177/1745691618804166.

a growth mindset can drive significant improvements in learning, especially when paired with a stress-can-enhance-performance mindset.[154] That last bit—that stress can help with growth—implies that challenge can improve your outcome, rather than limit you.

Maybe you are wondering: If stress is so great for learning, then why aren't we all great learners? We all certainly feel stress. I think it's because for most of us, the feeling we have when we fail at doing something difficult overrides the desire to keep striving. That is one reason I continually share ways for real-time autonomic self-regulation, like the physiological sigh described in chapter 3. These protocols are designed to lower stress not only to help people feel calm, but also to expand people's sense of what's possible to learn. If we have the tools to control stress, we can seek out more challenge.

No matter how well we control our stress, however, we are still left with the challenge of how to activate a growth mindset and stress-can-enhance-performance mindset. In

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[154] J. L. Burnette et al., "A Systematic Review and Meta-Analysis of Growth Mindset Interventions: For Whom, How, and Why Might Such Interventions Work?," Psychological Bulletin 149, no. 3–4 (2023): 174–205, https://doi.org/10.1037/bul0000368; Victoria F. Sisk et al., "To What Extent and Under Which Circumstances Are Growth Mind-Sets Important to Academic Achievement? Two Meta-Analyses," Psychological Science 29, no. 4 (2018): 549–71, https://doi.org/10.1177/0956797617739704; A. J. Crum et al., "Rethinking Stress: The Role of Mindsets in Determining the Stress Response," Journal of Personality and Social Psychology 104, no. 4 (2013): 716–33, https://doi.org/10.1037/a0031201.

other words: How do we actually get motivated? Should we use self-talk? Watch an inspirational clip on YouTube? Fantasize about achieving our goal? Scare ourselves with the fear of not achieving it? How many milligrams of fear should we take? Just kidding. Sort of.

The answer starts, I believe, with swapping the word motivation with a state of urgency. Here is why:

- A state of urgency is less prone to parsing than motivation, which immediately raises questions about context. Friedrich Nietzsche offered powerful wisdom when he wrote, “Hat man sein warum? des Lebens, so verträgt man sich fast mit jedem wie?”[155] Some readers may be more familiar with the English translation made famous by Austrian neurologist, psychiatrist, and Holocaust survivor Viktor Frankl: “He who has a why to live for can bear with almost any how.”[156]
- Sometimes the why is clear. But what about when it isn’t? What about the many aspects of learning and life that require persistent effort and for which the why, the deeper meaning and larger purpose, is unclear? Often the why only becomes clear during, or even after, the how.

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[155] Friedrich Nietzsche, The Case of Wagner, Nietzsche Contra Wagner, The Twilight of the Idols, The Antichrist, trans. Thomas Common (H. Henry and Company, 1896), 100.

[156] Viktor E. Frankl, Man’s Search for Meaning, trans. Ilse Lasch (Rider, 2004), 84.

Maybe a person is forced to take a course they hate in order to get a degree. Once they complete the course, however, they may realize the new opportunities have been made accessible to them only because of the material they learned in the course; it wasn't just a means to an end after all. A state of urgency doesn't require you to have all the answers yet.

- A state of urgency stimulates catecholamine release, which further increases sympathetic tone, which therefore increases focus.[157] This helps dim distractions, both internal and external, which improves performance… and increases your state of urgency.[158] It's a positive feedback loop.
- It's obvious why a sense of urgency would help you maximize the number of repetitions that you can do in a given time frame. That's huge because genuine attempts are the fundamental unit of all forms of learning.
- It's specific enough that we know what to aim for, experientially, but broad in the sense that it applies to all activities and pursuits. Motivation drives you to try something, a state of urgency keeps you practicing.

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[157] Chu et al., "Physiology, Stress Reaction."

[158] Andrew Westbrook and Todd S. Braver, "Dopamine Does Double Duty in Motivating Cognitive Effort," Neuron 89, no. 4 (2016): 695–710, https://doi.org/10.1016/j.neuron.2015.12.029.

So now the question is not, “How do we get motivated?” It’s “How do we create a state of urgency?”

The answer is mental contrasting, which is a tool based on the research of Gabriele Oettingen, PhD, a professor at New York University and the University of Hamburg.[159] The term is the result of her deep dive into understanding two psychological tools long thought to be helpful to this topic: positive thinking and visualization. The terms are likely familiar to you. Positive thinking refers to the practice of deliberately holding an optimistic view about an outcome as a way to help that hoped-for result come true. Visualization refers to the practice of imagining a desired future outcome, again as a way to help that result come true. Research had shown that a positive mindset can have beneficial effects on health and well-being.[160]

But as Oettingen and others have discovered, although positive thinking and visualization are great, they often undermine motivation just as easily as they can increase it. And get this: Too much positive thinking about goal

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[159] Gabriele Oettingen et al., “Mental Contrasting and Goal Commitment: The Mediating Role of Energization,” Personality and Social Psychology Bulletin 35, no. 5 (2009): 608–22, https://doi.org/10.1177/0146167208330856.

[160] Angel D. Prabakar, “The Power of Thought: The Role of Psychological Attentiveness and Emotional Support in Patient Trajectories,” Yale Journal of Biology and Medicine 97, no. 3 (2024): 335–47, https://doi.org/10.59249/CPTG1770.

outcomes undermines performance.[161] Apparently, visualization causes people's brain reward mechanisms to get sufficiently activated, so they feel less energy to continue to strive for the goal.[162] And positive thinking provides a big but transient boost in motivation.

All that is consistent with what we know about dopamine in the context of goal seeking and rewards. As Anna Lembke, an expert on dopamine, put it, dopamine is about wanting, not about having.[163] That's another downside of positive thinking: It activates the "want" circuitries but not the "have" ones. Anna further pointed out that the neural circuits underlying motivation evolved in times of scarcity, where effort almost always preceded reward—think hunting, gathering, seeking shelter, finding a mate, and so on. Because of the abundance of stimulation in modern life, which was designed to satisfy and even exploit our dopamine circuitry, most of us have recalibrated our motivation and reward circuits to expect more reward from

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[161] Gabriele Oettingen and Doris Mayer, "The Motivating Function of Thinking About the Future: Expectations Versus Fantasies," Journal of Personality and Social Psychology 83, no. 5 (2002): 1198–1212, https://pubmed.ncbi.nlm.nih.gov/12416922/.

[162] Gabriele Oettingen, "Future Thought and Behaviour Change," European Review of Social Psychology 23, no. 1 (2012): 1–63, https://doi.org/10.1080/10463283.2011.643698.

[163] Anne Lembke, Dopamine Nation: Finding Balance in the Age of Indulgence (Penguin, 2021).

less effort.[164] And while Anna's work has mainly focused on addictive and compulsive behaviors, we've all seen this play out for goals too.[165] Too much reward prior to effort, or too much reward without effort, undermines future effort.

A friend of mine, an MD turned successful startup investor, warned me to be careful about telling too many people about a big goal. Most will respond, he explained, by saying "You would be great at that, go for it!" Unless you have a lot at stake, the reward of all that positive feedback can undermine your drive to actually pursue the goal. I asked him if he thought it would be better if we responded instead by saying, "Don't try, you'll never succeed!" "I don't think that helps either," he replied.

He was right. Oettingen found that if people focus too much on the challenge they'll likely face, that too prevents them from pursuing goals, at least in the long run.

So much for just "Focus on the journey not the destination."

We can thus all be grateful to Oettingen for carrying out experiments that tested various common motivational tools, alone and in combination. She also explored their efficacy for helping people achieve different types of goals: learning

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[164] John P. Grogan et al., "Dopamine Promotes Instrumental Motivation, but Reduces Reward-Related Vigour," eLife 9 (October 1, 2020): e58321, https://doi.org/10.7554/eLife.58321.

[165] Lembke, Dopamine Nation.

cognitive skills, learning physical skills, making better food choices, exercising more, and on and on.[166] She eventually discovered that it is the contrast between visualizing the goal and thinking about the obstacles to achieve it that creates a state of heightened physical and mental energy, and that state is what leads us to be more likely to start—and to persist.[167]

In other words, Oettingen figured out how people can activate a state of urgency. She calls the approach mental contrasting, and she calls the way to implement it WOOP, an acronym that stands for Wish, Outcome, Obstacles, Plan.[168] The four steps of WOOP can be done in your head, but it works best if you record your answers. Try writing them down, or hitting the record button on your phone. It works for both short- and long-term goals.

WISH: Name the big, overall goal. What do you want to happen?

For example:

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[166] Gabriele Oettingen's Publications, https://wp.nyu.edu/motivationlab/publications/gabriele-oettingen/.

[167] Andreas Kappes et al., "Mental Contrasting Instigates Goal Pursuit by Linking Obstacles of Reality with Instrumental Behavior," Journal of Experimental Social Psychology 48, no. 4 (2012): 811–18, https://doi.org/10.1016/j.jesp.2012.02.002.

[168] Gabriele Oettingen, Rethinking Positive Thinking: Inside the New Science of Motivation (Current, 2015).

- “I want to learn guitar well enough to play with a band live in front of a thousand people.”
- “I want to get a PhD in neuroscience from a top program.”
- “I want to learn to speak fluent German.”
- “I want to paint well enough to make a living selling my art.”

It takes only a few seconds to complete the W step in WOOP. You don’t need to include a lot of detail, but some helps. For instance, “I want to be happy and successful” is too general. Instead of focusing on what you want to feel, name skills, credentials, and proficiencies that you’re hoping to achieve.

OUTCOME: Here is where you think about what you want to feel from having achieved the goal.

For example:

- “I will experience the thrill of performing live with other great musicians and knowing that I’m giving people the live music experience I know and love so much too.”
- “I will feel the freedom and excitement of being able to work for any number of different companies exploring for cures to neurodegenerative diseases.”
- “I will enjoy going to Germany to visit my spouse’s family, be able to relate to them better, and maybe even

work and start a life there someday. And I can help our kids become bilingual."

- "I will finally get to indulge my desire to be creative. And I can sleep in late and stay up late painting while listening to music, which is what I always wanted!"

You'll notice this outcome step is aspirational—you're encouraged to daydream about how you want to feel. I gave short examples here but feel free to expand on the many things you look forward to feeling. The reality check is next.

OBSTACLES: Get clear on the obstacles you anticipate en route to your goal.

Be hyperrealistic. Record the roadblocks you might face. Include how you might feel as you do. People's temperaments vary, but you know yourself—so be honest about how you'll react when things don't go to plan.

For example:

- "I tend to be really tired from work at the end of the day. I'm going to feel frustrated when I am too tired to practice guitar or when traffic makes me late to guitar lessons."
- "I will miss the chance to attend weddings and go on fun vacations with my friends because I'll be in the lab doing experiments or studying. I won't have much money because PhD students don't get paid much."

- “It will be a long time before I am truly fluent in German because I learn really slowly. And writing in German is even more difficult than speaking it, so I will get discouraged a lot.”
- “The art I make is not likely to appeal to a lot of people because I like melancholy themes and darker tones, and most people seem to buy art that is happy and playful.”

Again, feel free to expand as much as you want to for this step.

PLAN: Come up with if-then statements to address the obstacles you anticipated.

These are action steps.

For example:

- “On days when I’m too tired to practice guitar after work, I will go to sleep early and get up thirty minutes earlier than usual the next day. That way I can get fifteen to twenty minutes of practice done in the morning. My mind is much sharper then anyway. Playing in the morning also makes me feel good for hours, so maybe work might not feel so exhausting. If I’m late to guitar practice because of traffic, then I will do my best to stay calm and practice in my head on the way there. I will add a few minutes of practice in my car after the lesson to really drill in what I learned.”

- “When I can’t attend fun events because of time or money restrictions, I will remember that having a PhD in neuroscience puts me at a unique advantage for job choices and long-term financial security because there is so much still to discover about nervous system diseases… and we still don’t have cures to most all of them. Also, a lot of my friends who make more money than me now are bored with their jobs or have already maximized their earning potential. I will be able to consult and/or invest in biotech companies because I really understand where the field is and where it might grow. Also, I will remember that I love doing experiments.”
- “If I feel frustrated at my slow progress learning German, I will remind myself that the frustration is a signal my brain is getting the message to change, and that a lot of learning happens in bursts; it’s rarely linear. I will ask my spouse (who is fluent) if we can do an only-in-German dinner once per week as a family, so I and the kids can practice. I will use AI to generate some free German language quizzes for me that I can listen to when I’m alone in the car.”
- “I will remind myself that the best art has always come from people who really care about the work they are doing, and that I really feel and believe in the messages that motivate the art I create. I’ll set up a website and social media platform so I can connect with other artists

with similar tastes, and learn from them about marketing my work."

You may have noticed what happened there. Mental contrasting quickly moves you from idealizing and making aspirational statements about goals and feelings to concrete action steps… and, more importantly, clarity about how you will deal with setbacks. It customizes the process of getting motivated. If you're optimistic and unflappable and start new projects all the time or if you're pessimistic and tend to talk yourself out of things or if it depends on the circumstances or the time of day or the goal, mental contrasting still works. It also serves as a strategy setting session. It's not a pep talk. It's a preparation exercise that has been proven to help people get started and persist in pursuit of their goals. It's also zero cost, and takes around ten minutes at the outset. It's a real gift to us all, and we have Oettingen, along with the students and postdocs that ran the experiments, to thank for it.

In My Experience

I used mental contrasting for a goal that I had been procrastinating on forever.

I followed the WOOP process. I typed out the WOOP answers into a word processing document and hit Save. That took about ten minutes.

And it worked spectacularly well for me. It somehow cleared away the laziness and analysis paralysis and all the other stuff that had been holding me back, and I enjoyed two solid work sessions that very day, and then two months of steady effort toward the goal thereafter, even though I was in my busiest season of work and I had many other commitments.

When I started feeling a bit unmotivated again, I reread the document. I made some edits and additions based on new changes in life. That only took about five minutes. I now do mental contrasting for learning goals, for work goals, for fitness goals, and for personal goals too. I reread and update the WOOP documents whenever I am feeling a bit unmotivated. It's become reflexive. I don't share my WOOP answers with anyone, they are for me and for me only, but you can decide if you want to keep yours private or not.

The Mechanisms Behind Mental Contrasting

I was struck that mental contrasting seemed to increase my motivation across the board, not just for my specific goal. I procrastinate far less on everything now. I'm not aware of any data showing that doing mental contrasting rewires specific brain areas, but given what we know about the aMCC, I wonder if it might. Recall how Joe Parvizi conducted incredible experiments showing that if activity in the aMCC is increased, people feel an immediate surge in

mental energy and readiness to lean into challenges. Especially since every instance of learning—whether that's a new behavior or a new piece of information—is the consequence of plasticity and our aMCC is the brain structure most closely associated with tenacity.[169] It has been shown to increase in volume in successful dieters, people who push themselves to stick to a new exercise program, and other situations where thinking about and overcoming a challenge is called for. Parvizi discovered the role of the aMCC in 2013, and only recently has it been discussed in the larger field of neuroscience and media, so I doubt that Oettingen was aware of its potential role in WOOP. I look forward to a day soon when someone explores whether the aMCC is activated by going through the steps of the WOOP Protocol. In the meantime, I plan to keep using mental contrasting any time I need to generate a state of urgency for something, and I hope you'll consider it too.

Why Some People Will Avoid Mental Contrasting

This may surprise the dreamers and go-getters, but a lot of people are afraid to strategize around—let alone think

[169] Alexandra Touroutoglou et al., "The Tenacious Brain: How the Anterior Mid-Cingulate Contributes to Achieving Goals," Cortex 123 (February 2020): 12–29, https://doi.org/10.1016/j.cortex.2019.09.011.

about—their big goals. Some of that is cultural[170] but in my observation a lot of people are afraid to change, even if they are unhappy with their current circumstances. It's just not reflexive (and in many cases it feels aversive) for them to do so. Before you conclude that a resistance to change is just the way you're wired, consider the experiment where people preferred to get strong electrical shocks they knew were coming at specific times… versus weaker electrical shocks that arrived without warning.[171] The same general conclusion comes from studies of pain. People like what's predictable, even if it isn't good. That defies logic until we remember that our nervous systems are prediction machines that seek to minimize energy expenditure.[172] So, while mental contrasting is merely a mental exploration of a goal, I anticipate some people will shy away from it… because change is hard. I hope you don't and instead, go with the data. Put it to work for you. WOOP is a really cool tool to explore new things, and I'm grateful Gabriele Oettingen and colleagues came up with it. And learning to switch on a state of urgency to increase focus and achieve your goals and get

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170 Yuri Miyamoto et al., "Cultural Variation in the Focus on Goals Versus Processes of Actions," Personality and Social Psychology Bulletin 39, no. 6 (2013): 707–19, https://doi.org/10.1177/0146167213483579.

171 Archy O. de Berker et al., "Computations of Uncertainty Mediate Acute Stress Responses in Humans," Nature Communications 7 (2016): 10996, https://doi.org/10.1038/ncomms10996.

172 Abdullahi Ali et al., "Predictive Coding Is a Consequence of Energy Efficiency in Recurrent Neural Networks," Patterns 3, no. 12 (2022): 100639, https://doi.org/10.1016/j.patter.2022.100639.

things done in life is a valuable superpower, for SDAP and beyond.



LEARNING PROTOCOL

Embrace Errors to Improve Your Performance

WHAT TO DO

- Embrace the fact that errors are an essential trigger for learning. Errors that come from genuine attempts to learn something are the signal for plasticity to occur. This is true for all forms of skill development: cognitive, motor, musical, mathematical, et cetera, and why it's an essential step in SDAP.
- Strive to perform at a level of difficulty just slightly beyond your current ability. Studies show that the right level is such that it produces an error rate of 15 to 30 percent. The 30 percent level tends to work best for learning motor skills, whereas 15 percent is the approximate level to shoot for with cognitive skills.
- Don't make errors deliberately. That won't have the same result. Work on pushing yourself past what you think you can do and using your mistakes as learning tools.

HOW IT WORKS

If you can do something with proficiency, your nervous system won't change. The gap between your current level of ability (even if it's zero!) and the level you are striving for is what triggers learning and plasticity. No gap, no change.

Before I dive into this protocol, I need to make something clear. There are many forms of learning and plasticity that don't require making errors. Just look at what happens with children. As every parent knows, a toddler learns that ice cream is delicious in a matter of seconds, and never forgets it. They also learn that tablets and phones can display cartoons after seeing them only once. No errors needed. (By contrast, learning to walk, to speak, or to eat with utensils are all examples of learning and plasticity that require errors to eventually master.) Traumatic events, at any age, are also examples of error-independent learning and plasticity. So is the learning that comes from experiencing a welcome surprise. Let's say you drop by a friend's party expecting to say hello and then go home early, but you end up meeting someone you really like there. You can be sure you will remember their name, their job, et cetera. Should a great long-term relationship emerge from it, you'll no doubt maintain fond memories of the party, the friend who hosted it, and the events of the entire weekend when the party occurred.

You can thank dopamine for that. Dopamine helps us learn by linking our memories of prior actions, and the contexts in which they took place, with our sense of whether things went better or worse than we expected—a phenomenon known as reward prediction error.[173] It makes perfect sense why this evolved: to help us find and remember the paths we took to life-sustaining resources—food, water, shelter, potential mates, et cetera. Unwelcome surprises, even if they are not traumatic, shape our memories, help us learn, and guide our future actions too. Whereas positive events simultaneously increase dopamine and lower serotonin, negative events do the opposite. Say that instead of meeting a new friend at a party, you unexpectedly run into an old boss you hoped to never see again. That would shape your expectations and motivations around future gatherings at that house. It may even change your feelings about the friend—a phenomenon known to neuroscientists as context-dependent plasticity.[174]

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[173] Mitsuko Watabe-Uchida et al., "Neural Circuitry of Reward Prediction Error," Annual Review of Neuroscience 40 (July 25, 2017): 373–94, https://doi.org/10.1146/annurev-neuro-072116-031109.

[174] James B. Heald et al., "The Computational and Neural Bases of Context-Dependent Learning," Annual Review of Neuroscience 46 (2023): 233–58, https://doi.org/10.1146/annurev-neuro-092322-100402; Thomas M. Tzschentke, "Review on CPP: Measuring Reward with the Conditioned Place Preference (CPP) Paradigm: Update of the Last Decade," Addiction Biology 12, no. 3–4 (September 2007): 227–462, https://doi.org/10.1111/j.1369-1600.2007.00070.x.

Why We Need To Make Mistakes

You'll notice that none of the examples of learning I just described require that we make errors. Stuff happened, for better or worse, and learning occurred. It speaks to the many types of plasticity.

SDAP is different. We need to make errors. With errors often comes some amount of agitation, of limbic friction, which by now you know is a good thing because it increases the release of catecholamines that facilitate focus and plasticity. But also, perhaps more importantly, mistakes provide the signal that sets the size and type of neural circuit changes (and thus skill improvement) that follow learning attempts.

I'll use motor skills as an example but the same holds true for all types of skills. Let's say you want to learn tennis. You need to learn to serve the ball. At first, it's likely that you'll hit the ball into the net often. Each time you do, it's an error that you register as objective performance feedback. You adjust your serve accordingly. You may decide "that last serve was too hard, I'll use less force this time" or "perhaps I am gripping the racket too high on the handle, I'll hold it lower" or "I'm throwing the ball too high before I swing," and so on. Even if you have a tennis coach who gives you instructions, you are the only one who has access to the proprioceptive information associated with each serve

attempt—by that I mean the awareness of your body's movements. Your nervous system tracks each and every one of those, and compares it to the performance outcome. Some of that is conscious, and some is unconscious, working below your level of awareness.

Remember back in chapter 3, when we discussed what your nervous system does? As you may recall, whether you are asleep or awake, your nervous system is handling the basic stuff of survival, making sure that your heart keeps beating, your lungs keep expanding, your digestion keeps flowing, and so on. It is also letting you perceive the world around you. And, as we also discussed, your nervous system is also a prediction machine. It's constantly trying to make causal inferences: "If I do A, then B seems to happen immediately afterward… If I don't do X, then Y happens a few hours later…" When you are awake, your nervous system is running a ceaseless if-then analysis of you and your actions and feelings, thoughts and memories; of others and their actions and feelings, thoughts and memories; of the world around you, and on and on.

Here is the key thing to remember: Any time your nervous system figures out that one of your actions leads to a successful outcome, it starts to shift the pattern of neural firing that occurred from your conscious awareness into your implicit motor memory. It works to make the action

automatic. The rationale for that is obvious: Thinking and doing and making associations is metabolically demanding. As is rewiring brain circuitry.[175] The more your brain can make things automatic, the more energy it can devote to other things. This is especially true in the case of skill learning. With maintaining focus, moving into a state of urgency, making errors, and calibrating adjustments, we can all learn to execute new skills. Once we do, we don't need to think about each step. We simply activate performance of the skill. We serve the ball, play the chord, finish the math problem. Voila!

I acknowledge that your reaction to this point might be: "Duh!" But I said there was a key thing to remember. It will forever change the way you approach learning.

First, your brain doesn't work alone here. It needs your second brain. No, not your gut. I'm referring to the mini brain tucked right under your occipital lobe. If you touch the top of your neck where it meets the back of your skull, you're touching the area just above your cerebellum, aka your mini brain. Many people have heard of the cerebellum for its role in balance, but your cerebellum is also crucial for eye movements related to attention, for proprioceptive awareness, for speaking, for estimating time, for your sense

[175] Sebastian Rumpf et al., "Energy Metabolic Pathways in Neuronal Development and Function," Oxford Open Neuroscience 2 (March 21, 2023): kvad004, https://doi.org/10.1093/oons/kvad004.

of rhythm, and much more. Your cerebellum is also a key site of action for comparing what you are trying to do with what you're currently doing, and sending the rest of your nervous system the signals it needs to close the gap in performance.

I would love to devote a thousand pages to the function of the cerebellum in learning because it is that cool, but I'll keep it (relatively) brief. When you make errors in the context of trying to learn a new skill, you activate neurons in your cerebellum. The bigger the error, the larger the response from your cerebellum. Your cerebellum is kind of like that tennis coach in this regard, except unlike a coach that observes you and makes educated guesses as to what you need to do in order to improve, your cerebellum has full access to both your inner and outer workings: the angles of your limbs and joints, the speed and stretch of your muscles; it knows when you blinked relative to your maximum effort, your level of autonomic arousal, et cetera. Thus, your cerebellum is also uniquely suited to be your coach because of both the specificity and the interconnectedness of its wiring. Different cerebellar neurons respond to the movement of different parts of the body, even your eyes and lips and tongue. It also encodes how movement of one body area changes the movement of all the others. If you consciously adjust the angle of your elbow and it improves your serve, your cerebellum registers the many other

changes that accompanied the elbow adjustment and then makes sure they become part of your serve “motor program” going forward. If your elbow angle adjustment makes things worse, your cerebellum registers that too. With repeated effort, your brain eventually stamps down the exact motor program needed to serve the ball properly.

I wish we all learned about the role of the cerebellum in learning when we were kids. Instead, we watched Star Wars, Harry Potter, X-Men, and Marvel movies, et cetera, which are all wonderful stories, but sadly also embed the false idea in us that skills are things that just need to be activated, as if by magic. Of course, that’s fiction. The real-life superpower all of us possess from birth is our ability to focus and learn. We can progress from unskilled to skilled, and from skilled to mastery. Your cerebellum, and the larger neural circuit it participates in, is the coach that guides the entire progression. If that were the message, I believe the frustration that comes from errors would take on a whole new meaning. We would (and should) view errors and the agitation that come from genuine attempts to learn as something to seek out, not to avoid. Errorful learning, as some researchers call it, appears to be the most efficient approach to acquiring new skills.[176]

[176] Preeti Rishi et al., “Errorless and Errorful Learning in People with Aphasia Across Novel-Object Pairing and Word Retrieval Tasks,” Aphasiology 39, no. 10 (2025): 1381–1413, https://doi.org/10.1080/02687038.2024.2432022; Josie Briscoe et al., “Errorful Learning Improves Recognition Memory for New

The Goldilocks Zone for Errors

You may be asking, “How many errors are needed? And what kind?” In a study published in 2019, researchers set out to determine the optimal level of difficulty—the Goldilocks zone for learning new skills.[177] They calculated the ideal rate of error using several learning algorithms and found that training progressed much faster when the level of difficulty set was associated with an accuracy rate of about 85 percent (an error rate of 15 percent). A few additional insights emerged from this study. First, when something is too hard such that 50 percent of attempts were unsuccessful, the brain can’t seem to capture the right strategy for neural circuit modification. It was much better, the researchers found, to continually adjust the training level to maintain the approximately 15 percent error rate. This was a quantitative

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Vocabulary for People Living with Memory and Dysexecutive Impairment Following Brain Injury,” Neuropsychological Rehabilitation 34, no. 7 (2024): 974–1004, https://doi.org/10.1080/09602011.2023.2259017; Steven C. Pan and Faria Sana, “Pretesting Versus Posttesting: Comparing the Pedagogical Benefits of Errorful Generation and Retrieval Practice,” Journal of Experimental Psychology: Applied 27, no. 2 (2021): 237–52, https://doi.org/10.1037/xap0000345; Yeray Mera et al., “Unraveling the Benefits of Experiencing Errors During Learning: Definition, Modulating Factors, and Explanatory Theories,” Psychonomic Bulletin & Review 29 (2022): 753–65, https://doi.org/10.3758/s13423-021-02022-8; Naser Al-Fawakhiri et al., “Evidence of an Optimal Error Rate for Motor Skill Learning,” preprint, bioRxiv (July 2023): 549705, https://doi.org/10.1101/2023.07.19.549705.

177 Robert C. Wilson et al., “The Eighty-Five Percent Rule for Optimal Learning,” Nature Communications 10, article no. 4646 (2019), https://doi.org/10.1038/s41467-019-12552-4.

modeling study, meaning the findings are based on a simulation, but it was aligned with studies of visual-perception-based decision-making and motor learning.

It works for cognitive tasks too. For example, students who are allowed to figure out how to solve complex math problems on their own—and therefore can make mistakes—tend to do better than students who are told the right way to do it from the start.[178] And we seem to know this intuitively. In one study, researchers at Columbia University asked fifty-two people to play a series of computer games.[179] They were free to choose from a range of games, some of which were easy, some harder, and some impossible to learn. As the task progressed, participants chose increasingly difficult games and preferred to repeat the challenging games rather than repeat the easy ones.

Remember the 15 percent "rule" is an approximate measure. There's no need to become obsessed with maintaining this exact error rate. To put it in simple terms, you want to work

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[178] Manu Kapur and Katerine Bielaczyc, "Designing for Productive Failure," Journal of the Learning Sciences 21, no. 1 (2012): 45–83, https://doi.org/10.1080/10508406.2011.591717.

[179] Adrien F. Baranes et al., "The Effects of Task Difficulty, Novelty and the Size of the Search Space on Intrinsically Motivated Exploration," Frontiers in Neuroscience 8 (October 13, 2014): 317, https://doi.org/10.3389/fnins.2014.00317.

at a level at which you're getting things right much of the time, but also wrong some of the time.

This insight can be used in education as well. Given the relationship between errors and plasticity, and the pivotal role of the cerebellum in that process, teachers should consider reserving verbal feedback to students in cases where the students thought they got something right but actually didn't. Otherwise, it's better to give students the opportunity to register their own errors.

The moment immediately following making a mistake is critical for learning. Negative reactions typically occur within 100 milliseconds,[180] but the sympathetic arousal lasts longer and provides the milieu for enhanced focus and plasticity as we have discussed many times now. Which means our systems are primed for learning in those immediate post-error moments. This internal state improves the effectiveness of the whole learning bout. Studies of motor learning show that within-trial errors and across-trial errors both provide valuable signals for plasticity and learning.[181] If you leave a language class or tennis lesson or

[180] Hans S. Schroder et al., "Neural Evidence for Enhanced Attention to Mistakes Among School-Aged Children with a Growth Mindset," Developmental Cognitive Neuroscience 24 (April 2017): 42–50, https://doi.org/10.1016/j.dcn.2017.01.004.

[181] Scott T. Albert and Reza Shadmehr, "The Neural Feedback Response to Error as a Teaching Signal for the Motor Learning System," Journal of Neuroscience 36, no. 17 (2016): 4832–45, https://doi.org/10.1523/JNEUROSCI.0159-16.2016; Dominique L. Pritchett and

other learning bout thinking, “Darn, I couldn’t seem to get that right on that one part,” well, congratulations! The learning process set in motion by within- and across-trial errors was triggered. You’ll no doubt come back better next time… but only if you do what’s in the next protocol.



LEARNING PROTOCOL

Prioritize Sleep the Night After a Learning Bout

WHAT TO DO

- Get adequate sleep the night after a learning bout. That’s when plasticity actually occurs, and why it’s an essential step in the SDAP process.
- Enhance plasticity by practicing ten to thirty minutes of NSDR at any point after a learning bout. Done prior, as described in Protocol 1 of this chapter, it will increase your mental and physical vigor.

HOW IT WORKS

During sleep, our brains rapidly replay what we tried to learn during the day, including what errors to avoid in the future. Brain imaging and electrophysiological recordings show that

Megan R. Carey, “A Matter of Trial and Error for Motor Learning,” Trends in Neurosciences 37, no. 9 (2014): 465–66, https://doi.org/10.1016/j.tins.2014.08.001.

the very same neural firing sequences that occurred during learning bouts occur again at that time.[182] If the patterns of brain activity in sleep are disrupted, we don't learn as well. Sleep isn't just a time for consolidation, it's also when we build schemas—which are extractions of common themes of movement or ideas that let our brain store information in a more efficient way. They serve as patterns that become part of our foundational understanding of how the world works.

Evidence points to REM sleep in the first night after a learning bout as being the most crucial for the learning consolidation process, a phenomenon known as the night after learning effect, but depending on the skill you're trying to master, deep slow wave sleep may matter more.[183] I wouldn't parse things that finely. Rather, do your best to get adequate (for you) sleep every night, but especially in times of striving to master new information and skills. I define adequate sleep in chapter 1, and give you lots of tools to use to achieve it. If you suffer a poor night's sleep, however, don't make things worse by fretting about the learning lost from it. Do NSDR the next morning, and get your sleep schedule back on track that night.

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[182] Jean-Baptiste Eichenlaub et al., "Replay of Learned Neural Firing Sequences During Rest in Human Motor Cortex," Cell Reports 31, no. 5 (2020): 107581, https://doi.org/10.1016/j.celrep.2020.107581.

[183] Matthew P. Walker, "The Role of Slow Wave Sleep in Memory Processing," Journal of Clinical Sleep Medicine 5, no. S2 (2009): S20–26, https://pubmed.ncbi.nlm.nih.gov/19998871/.

5

LEARNING PROTOCOL

Incorporate Micro Rests into Your Learning Bouts

WHAT TO DO

- Pause now and then between repetitions as you practice a new motor or cognitive skill. These waking micro rests have been shown to accelerate learning by increasing repetition density at the level of neural circuit activation. Short breaks of five to thirty seconds during a learning bout in which you do nothing at all (meaning, no checking your phone!) can significantly accelerate motor learning.
- Short breaks of five to ten minutes during a learning bout focused on cognitive learning can do the same. These micro rests induce ten to twenty times accelerated replay of the neural circuit firing pattern related to what you are trying to learn.

HOW IT WORKS

We try to perform a new skill and we fail, and fail again, and get close, but fail again. We get a few nights' adequate sleep, and when we come back for our next attempt, somehow we've improved.

The previous protocol explained how sleep reinforces neural circuit firing patterns that allow our brains to integrate new

skills. But recent studies show that a waking micro rest can increase consolidation on fast timescales too—meaning in just seconds to minutes, and even during the learning bout.

The observation that people learn faster if they pause from time to time between repetitions goes back to the late 1800s, when the German psychologist Hermann Ebbinghaus measured his ability to learn new material.[184] He saw that spaced repetitions led to faster and more stable learning than continuous repetition. That was self-experimentation, but many years later, in a study led by National Institutes of Health physiologist Marlene Bönstrup, twenty-seven participants were asked to practice several keyboard-based motor skills, tasks that were well established in the research of memory and learning.[185] Participants had to tap out a sequence of five items using their nondominant hand. They were given thirty-six tries to learn the sequence, with each attempt lasting ten seconds followed by ten seconds of rest. Performance was measured according to both speed and accuracy.

At first, Bönstrup and her colleagues examined the learning curve across all thirty-six attempts of a given sequence. They

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[184] Jaap M. J. Murre and Joeri Dros, "Replication and Analysis of Ebbinghaus' Forgetting Curve," PLOS One 10, no. 7 (2015): e0120644, https://doi.org/10.1371/journal.pone.0120644.

[185] Marlene Bönstrup et al., "A Rapid Form of Offline Consolidation in Skill Learning," Current Biology 29, no. 8 (2019): 1346–51.e4, https://doi.org/10.1016/j.cub.2019.02.049.

could see the performance improved rapidly, and then petered out at around the eleventh cycle. That's a familiar pattern. We often get better at a new skill fast and then reach a ceiling when further improvements become harder.

But the researchers noticed something interesting. All along, they had been monitoring the participants' brain waves. During very brief periods of rest between repetitions—lasting about ten seconds—the patterns of brain activity showed a unique signature. They tended to occur in regions connected to our ability to plan our movements. The researchers hypothesized that some form of additional rehearsal was occurring in their brains during the micro-rest periods, and it might be important for plasticity. That mapped well onto Ebbinghaus's observation. A second study, carried out in 2020 by Ethan Buch and others, found that micro rests induce rapid plasticity and learning, most likely because the same neural circuits active during the attempt to perform the motor skill get replayed during the micro rest in between repetitions, but at about twenty-fold the speed and forward and backward.[186] In other words, waking micro rests increase repetition density. This wakeful consolidation seems to allow you to integrate the new motor sequence with prior learned sequences. Of course, it's during

[186] Ethan R. Buch et al., "Consolidation of Human Skill Linked to Waking Hippocampo-Neocortical Replay," Cell Reports 35, no. 10 (2021): 109193, https://doi.org/10.1016/j.celrep.2021.109193.

sleep when our brains really engrave the circuits carrying newly learned information. Nonetheless micro rests of ten to thirty seconds during motor learning bouts, and five to ten minutes during cognitive learning bouts, have been shown to improve many different types of skills.[187]

You may be wondering what to do during the micro rests, and how often to take them. For a learning bout focused on motor learning, the key is to simply pause and not do another rep for five to ten seconds. A bit longer, say fifteen or seventeen or twenty seconds is not a problem, so don't obsess about the exact duration. Just pause for a short time, and then continue. You can draw a bit or just sit there—which sounds boring and might make you feel lazy, but the data say you are learning faster. Yes, you can get up and walk around, use the bathroom, et cetera. Just stay off your phone, web browser, or email…!

Introduce micro rests somewhat at random. How often you do them depends on the complexity and the duration of a repetition. Practicing serving a tennis ball is very different, for instance, from practicing an entire song on the piano. I suggest you start the learning bout by doing continuous repetitions one after the other, and when you start to make

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[187] Patricia Albulescu et al., " 'Give Me a Break!' A Systematic Review and Meta-Analysis on the Efficacy of Micro-Breaks for Increasing Well-Being and Performance," PLOS One 17, no. 8 (2022): e0272460, https://doi.org/10.1371/journal.pone.0272460; Bönstrup et al., "Rapid Form of Offline Consolidation."

errors, introduce a pause every third or fifth trial, then go back to continuous repetitions, and then pause after every error or every other or every third error. If you want, you could set a chime to go off at random intervals. A website that offers that feature for free is www.random.org/clock-times and there is another great, no-cost website at www.microbreaks.co/timer.html that lets you set the parameters, including (1) learning bout duration, (2) duration of micro-break, and (3) the tone you want to hear, et cetera. It's terrific for both cognitive and motor skill learning. (Note: I have no affiliation to it but I use it.)

Waking micro rests are a great tool for accelerating learning and plasticity that are based in behavioral and neuroscience-backed data. I suggest trying them.



LEARNING PROTOCOL

Increase Retention of New Skills and Information with Mental Rehearsal

WHAT TO DO

- Take one to five minutes to reflect on what you read or did during a learning bout at some point after the learning bout concludes. The key is to test yourself for what you do and do not remember, do or do not comprehend. That's why it's the final step of SDAP. For cognitive

skills and new information, this can be informal, done simply as a mental exercise, or more formal, such as by having a friend or ChatGPT test you for comprehension on the exact same material.

- For motor skills, think about where you were challenged and made errors.

HOW IT WORKS

All of learning is anti-forgetting.

On the one hand, that statement is another giant "DUH!" But through the lens of what we know about how humans learn and remember new information and skills, it's a very important concept to integrate.

The human brain can't remember all it senses and perceives. Even though neuromodulators can serve as a sort of highlighter pen for certain experiences, most information is forgotten. Neuroimaging studies have shown that for people who learn to juggle as adults, their brain circuits rewire to reflect the specific skills of juggling but those changes are quickly lost if they don't practice.[188] Both memories and new skills extinguish quickly, in days to weeks, unless they are locked in by rehearsal. The exception to this is, of course, the information related to very emotionally charged events,

[188] Elisabeth Wenger et al., "Expansion and Renormalization of Human Brain Structure During Skill Acquisition," Trends in Cognitive Sciences 21, no. 12 (2017): 930–39, https://doi.org/10.1016/j.tics.2017.09.008.

for reasons partially discussed earlier in this chapter. When adrenaline is very high, a second memory pathway involving our amygdala and other limbic-associated structures opens up and we remember more "snapshots" of whatever took place while in the high arousal state.[189]

Rehearsal in the form of self-testing turns out to be the best way to protect newly acquired skills and information against forgetting. A classic 2006 study from psychologists Henry Roediger and Jeffrey Karpicke compared repeated reading of a passage of text with one-time reading of the text combined with later mental reflection, and found that mental reflection was far superior when it came to both retention and comprehension of the material.[190] That's pretty wild if you think about it. People who spend less time with the actual material but attempt to review what they do and don't remember, and what they do and don't comprehend, can work with it more flexibly. That tells us that what we remember is not simply a matter of how much time we spend with information or in an experience, but that reflection

[189] Christa K. McIntyre et al., "Interacting Brain Systems Modulate Memory Consolidation," Neuroscience & Biobehavioral Reviews 36, no. 7 (2012): 1750–62, https://doi.org/10.1016/j.neubiorev.2011.11.001; K. C. Liang et al., "Involvement of Amygdala Pathways in the Influence of Post-Training Intra-Amygdala Norepinephrine and Peripheral Epinephrine on Memory Storage," Brain Research 508, no. 2 (1990): 225–33, https://doi.org/10.1016/0006-8993(90)90400-6.

[190] Henry L. Roediger III and Jeffrey D. Karpicke, "Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention," Psychological Science 17, no. 3 (2006): 249–55, https://doi.org/10.1111/j.1467-9280.2006.01693.x.

somehow reinforces and enhances the circuits carrying that information or experience.

This research has vast implications for educational settings, particularly for how best to study. We know that underlining and highlighting information only helps us remember it if we also later write down what was highlighted and underlined. It also means that testing—whether by our teacher or ourselves—is not just a form of evaluation to gauge our level of knowledge, it also represents a valuable "anti-forgetting" tool that helps us remember the information for months or years to come.[191]

The takeaway is clear: Tests work. If you're a teacher or coach, yes, this justifies pop and scheduled quizzes. It may help to test on recently learned material in exercises that

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[191] Roediger and Karpicke, "Test-Enhanced Learning"; Cynthia J. Brame and Rachel Biel, "Test-Enhanced Learning: The Potential for Testing to Promote Greater Learning in Undergraduate Science Courses," CBE—Life Sciences Education 14, no. 2 (June 2015), https://doi.org/10.1187/cbe.14-11-0208; Christopher A. Rowland, "The Effect of Testing Versus Restudy on Retention: A Meta-Analytic Review of the Testing Effect," Psychological Bulletin 140, no. 6 (2014): 1432–63, https://doi.org/10.1037/a0037559; Jeffrey D. Karpicke and Janell R. Blunt, "Retrieval Practice Produces More Learning than Elaborative Studying with Concept Mapping," Science 331, no. 6018 (January 2011): 772–75, https://doi.org/10.1126/science.1199327; Khalil A. Bin Abdulrahman et al., "Study Habits of Highly Effective Medical Students," Advances in Medical Education and Practice 12 (2021): 627–33, https://pmc.ncbi.nlm.nih.gov/articles/PMC8197661/pdf/amep-12-627.pdf; Sally Binks, "Testing Enhances Learning: A Review of the Literature," Journal of Professional Nursing 34, no. 3 (May–June 2018): 205–10, https://doi.org/10.1016/j.profnurs.2017.08.008.

count toward a grade. And, yes, don’t tell the students that until after the quiz.

For those without a teacher or coach navigating SDAP, reflect on what you learned, and most of all: self test! You can do so informally or formally. AI is a great tool for this. Ask ChatGPT to quiz you on a given set of material as a teacher would. Or simply take a few minutes and test your level of recall or physical skill on your own. Try to find where your ability drops off. It takes only a few minutes and is immensely valuable. You’ll learn faster and retain more of what you learn.